



10. 循环神经网络-1



建设单位： 大连理工大学
建设人： 孔雨秋



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01

PART ONE

引言

Introduction

Andrew Ng, Recurrent Neural Network

<https://www.bilibili.com/video/av88010803?p=1>



1

循环神经网络

→ 处理序列数据

语音识别



y

“The quick brown fox jumped over the lazy dog”

音乐生成



情感分类

“There is nothing to like in this movie.”



机器翻译

“Voulez-vous chanter avec moi?”



“Do you want to sing with me?”

视频行为识别



Running

命名实体识别

“Yesterday, Harry Potter met Hermione Granger.”



“Yesterday, **Harry Potter** met **Hermione Granger**.”



1

循环神经网络

→ 处理序列数据

- 触发字检测



Amazon Echo
(Alexa)



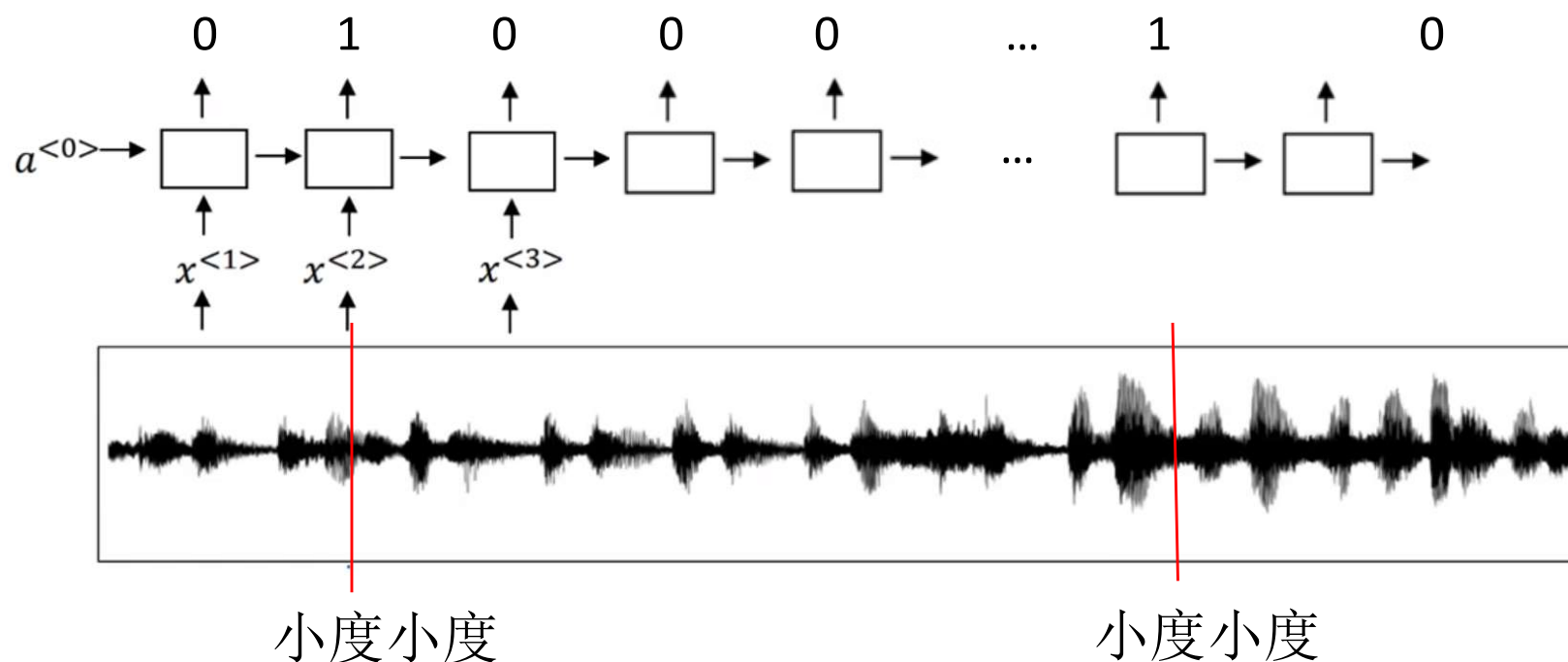
Baidu DuerOS
(xiaodunihao)



Apple Siri
(Hey Siri)



Google Home
(Okay Google)



PART TWO

基本循环神经网络

Basic RNN



1 基本循环神经网络

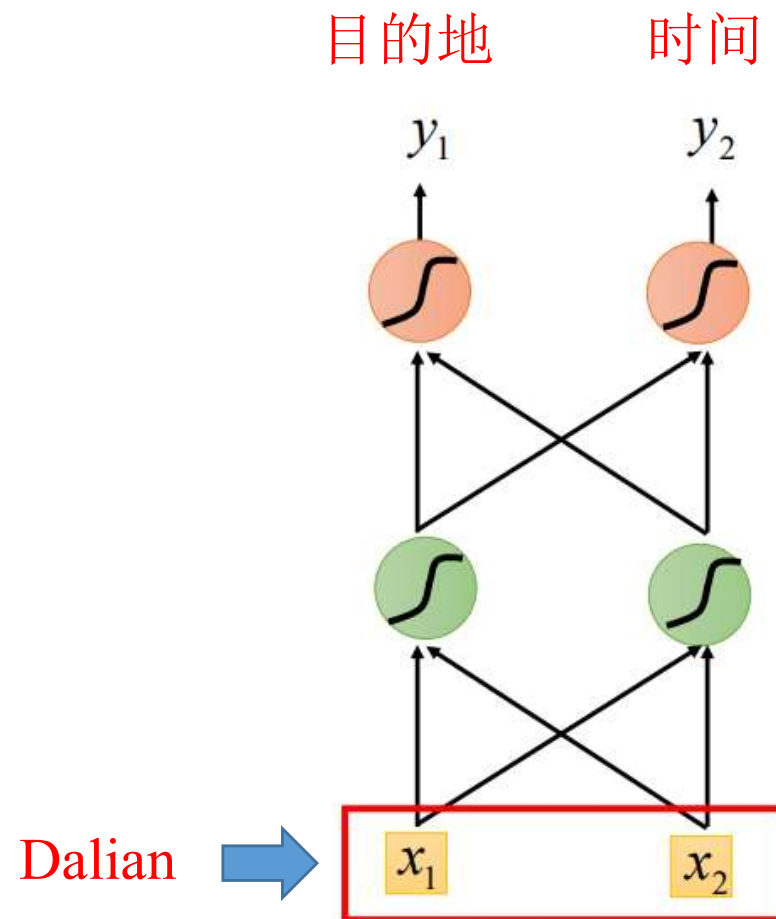
■ 前馈神经网络 v.s. 循环神经网络

Arrive **Dalian** on 1st December.

任务：预测目的地是哪？时间是？

输入：每次输入一个单词（特征表示）

输出：属于目的地或时间的概率



1 基本循环神经网络

■ 前馈神经网络 v.s. 循环神经网络

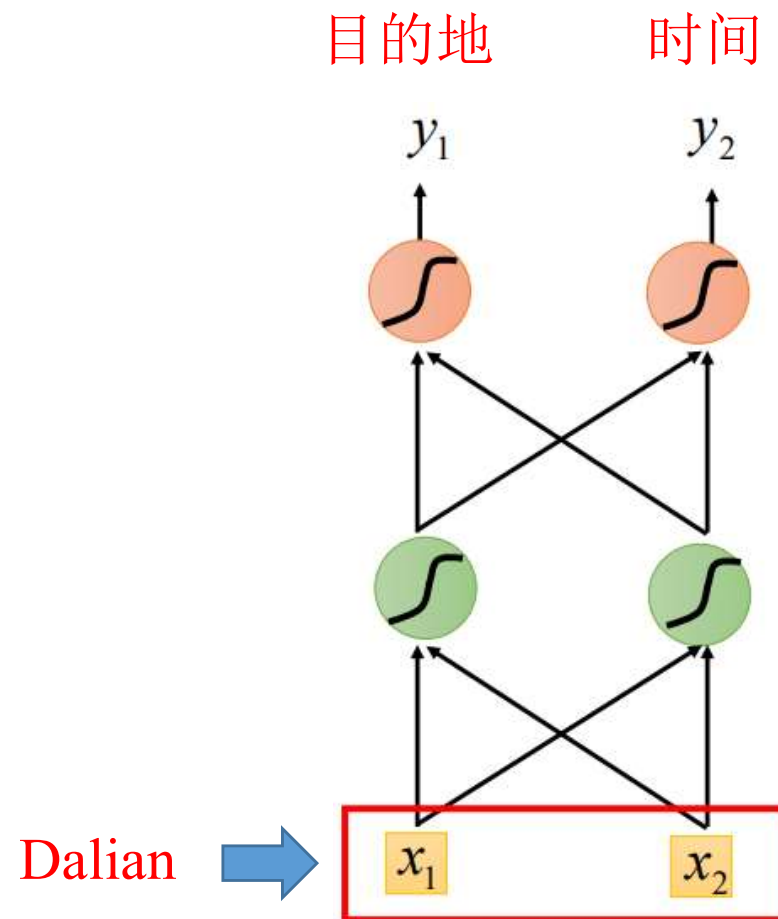
Arrive **Dalian** on 1st December.

任务：预测目的地是哪？时间是？

Arrive **Dalian** on 1st December. 目的地

Leave **Dalian** on 1st December. 出发地

前馈神经网络不考虑上下文信息，
输入是固定的，输出就是固定的。



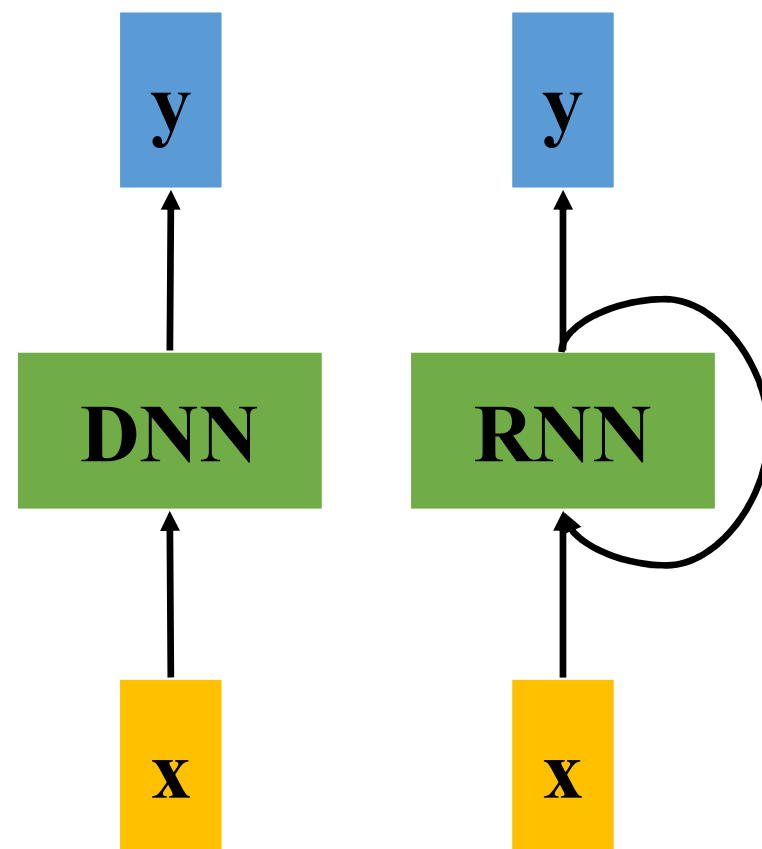
1 循环神经网络

■ 前馈神经网络 v.s. 循环神经网络

Arrive **Dalian** on 1st December. 目的地

Leave **Dalian** on 1st December. 出发地

- 神经网络需要记忆功能，捕获历史时刻的信息
- 将 t 时刻隐藏层的输出重新输入到隐藏层中，作为 $t+1$ 时刻的输入



1

命名实体识别

→ 符号

x

Harry Potter and Hermione Granger invented a new spell.

$T_x = 9$

$\mathbf{x}^{<1>} \quad \mathbf{x}^{<2>} \quad \mathbf{x}^{<3>} \quad \dots \quad \mathbf{x}^{<t>} \quad \dots \quad \mathbf{x}^{<9>}$

y

1 1 0 1 1 0 0 0 0

$T_y = 9$

$\mathbf{y}^{<1>} \quad \mathbf{y}^{<2>} \quad \mathbf{y}^{<3>} \quad \dots \quad \mathbf{y}^{<t>} \quad \dots \quad \mathbf{y}^{<9>}$

第*i*个样本

$\mathbf{X}^{(i)} = (\mathbf{x}^{<1>}, \mathbf{x}^{<2>}, \dots, \mathbf{x}^{<T_x^{(i)}>})$ $\mathbf{x}^{(i)<t>}$

$\mathbf{Y}^{(i)} = (\mathbf{y}^{<1>}, \mathbf{y}^{<2>}, \dots, \mathbf{y}^{<T_y^{(i)}>})$ $\mathbf{y}^{(i)<t>}$



1

命名实体识别



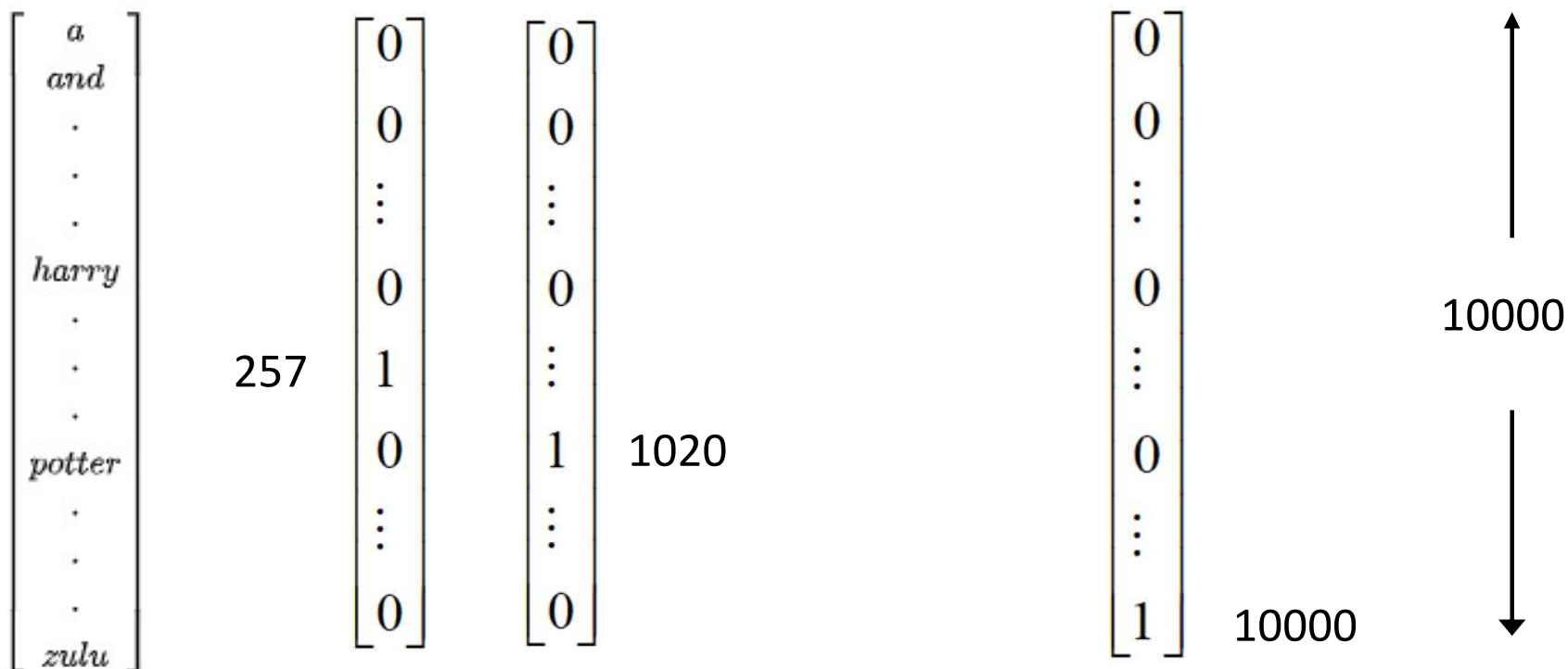
符号

x

Harry Potter and Hermione Granger invented a new spell.

$x^{<1>}$ $x^{<2>}$ $x^{<3>}$... $x^{<t>}$... $x^{<9>}$

Vocabulary



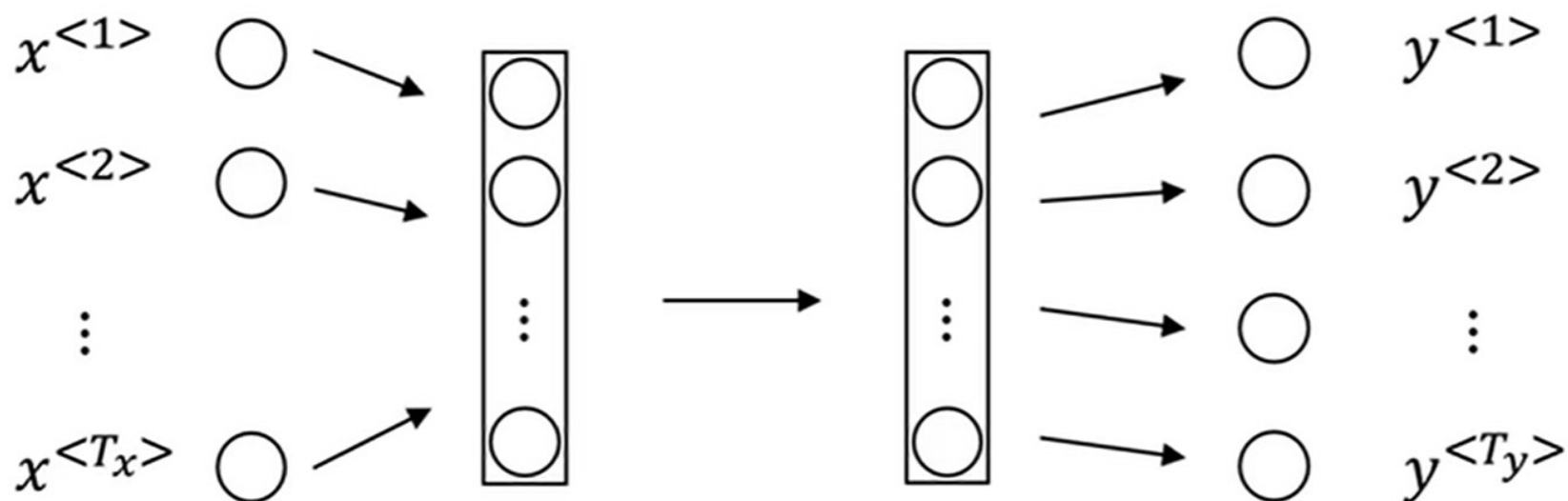
不在Vocabulary里的单词可用<UNK>标记



1

命名实体识别

- 为什么不能用一般的神经网络

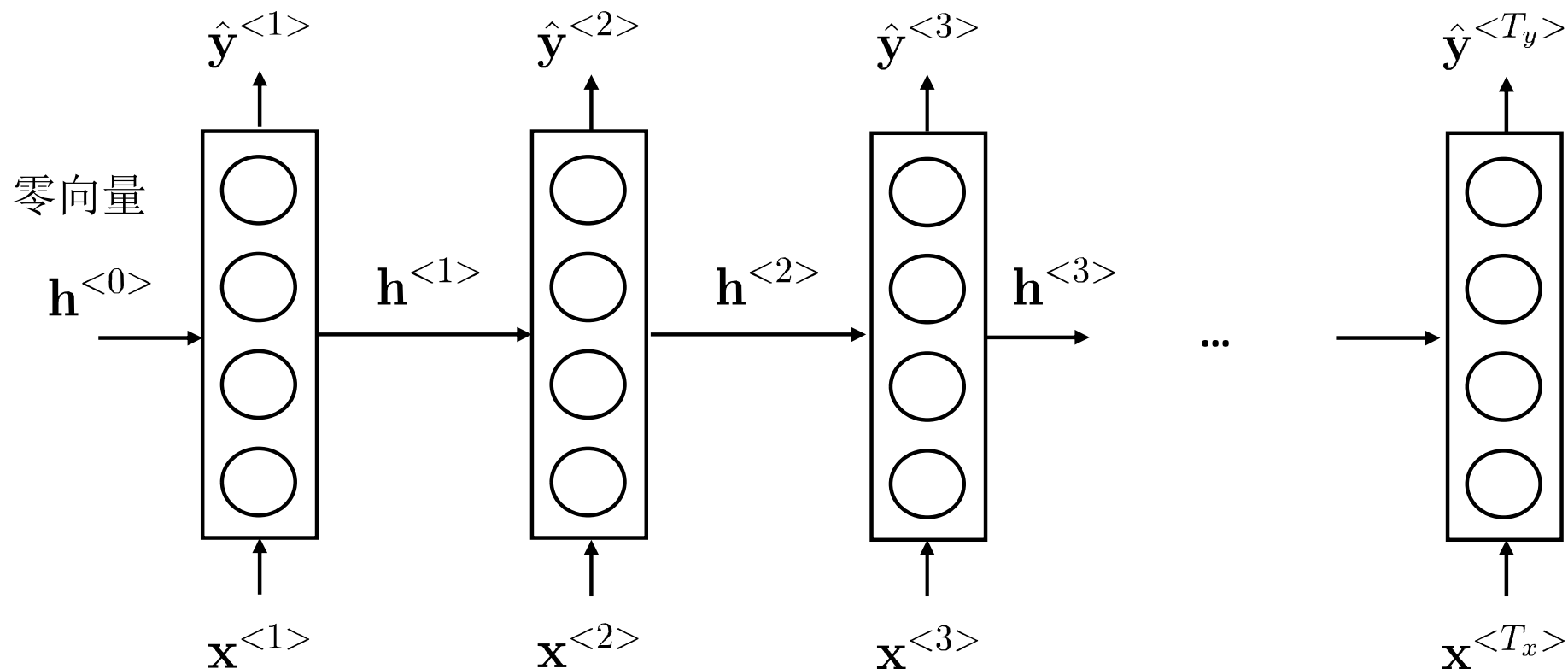


- 不同的训练样本，输入、输出序列的长度不同
- 在文本序列不同位置学习到的特征不共享

2

命名实体识别

→ 循环神经网络



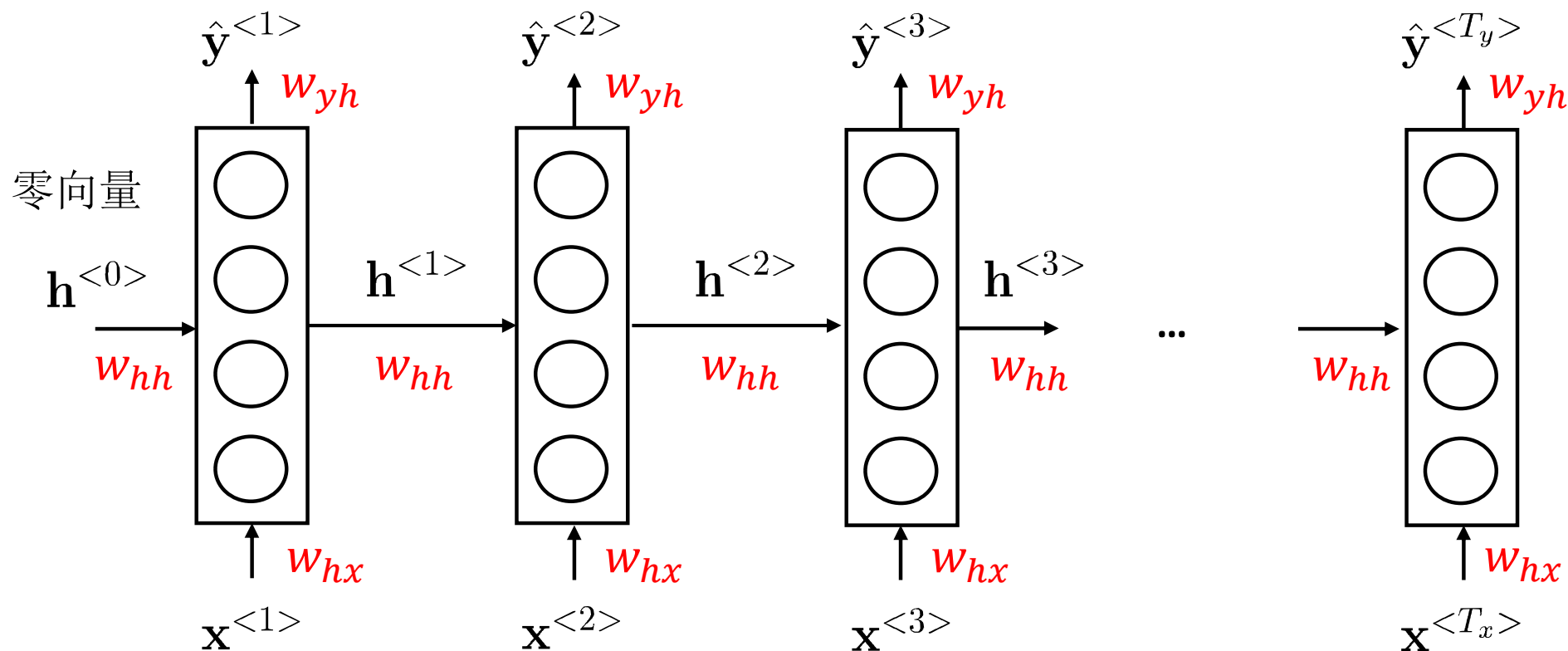
- 从左到右扫描序列数据
- 每个时间步共享参数



2

命名实体识别

→ 循环神经网络



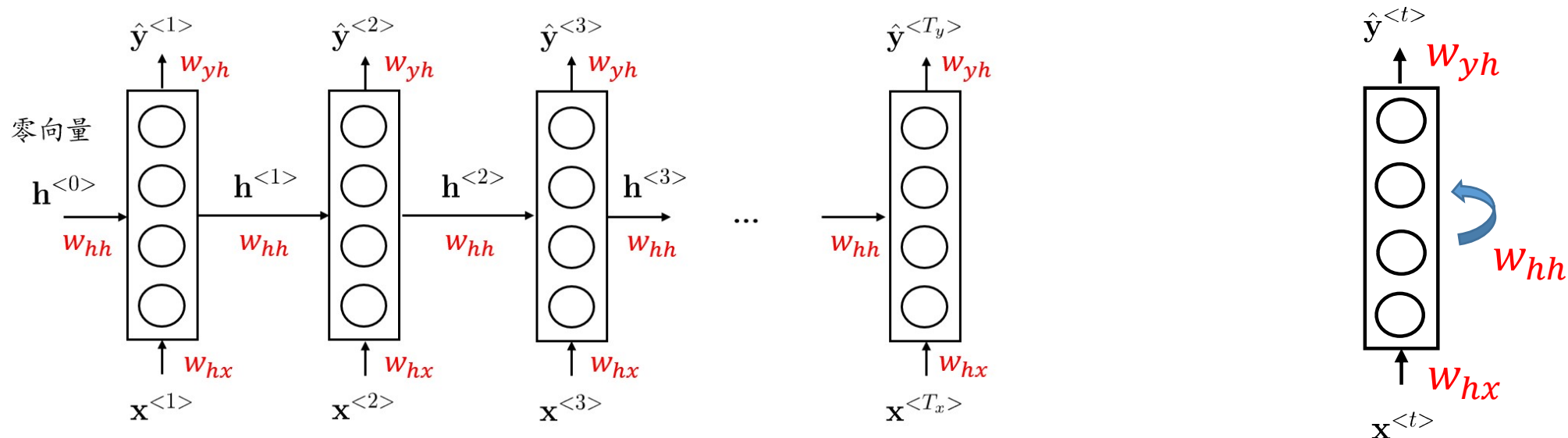
- 从左到右扫描序列数据
- 每个时间步共享参数



2

命名实体识别

→ 循环神经网络



$$h^{<0>} = 0 \quad \begin{cases} h^{<1>} = g(W_{hh}h^{<0>} + W_{hx}x^{<1>} + b_h) \\ \hat{y}^{<1>} = g_2(W_{yh}h^{<1>} + b_y) \end{cases}$$

激活函数: tanh

激活函数: sigmoid (二分类)
softmax (多分类)

$$\begin{cases} h^{<t>} = g(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b_h) \\ \hat{y}^{<t>} = g_2(W_{yh}h^{<t>} + b_y) \end{cases}$$



2

命名实体识别

→ 循环神经网络

$$\begin{cases} \mathbf{h}^{<t>} = g(\mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>} + \mathbf{b}_h) & \mathbf{h}^{<t>} = g(\mathbf{W}_h[\mathbf{h}^{<t-1>}, \mathbf{x}^{<t>}] + \mathbf{b}_h) \\ \hat{y}^{<t>} = g_2(\mathbf{W}_{yh}\mathbf{h}^{<t>} + \mathbf{b}_y) \end{cases}$$

(100,100) 100 (100,10000) 10000

$$\begin{cases} \mathbf{h}^{<t>} = g(\mathbf{W}_h[\mathbf{h}^{<t-1>}, \mathbf{x}^{<t>}] + \mathbf{b}_h) \\ \hat{y}^{<t>} = g_2(\mathbf{W}_y\mathbf{h}^{<t>} + \mathbf{b}_y) \end{cases}$$

$$\mathbf{W}_h = [\mathbf{W}_{hh}; \mathbf{W}_{hx}]$$

(100,10100) (100,100) (100,10000)

$$[\mathbf{h}^{<t-1>}, \mathbf{x}^{<t>}] = \begin{bmatrix} \mathbf{h}^{<t-1>} \\ \mathbf{x}^{<t>} \end{bmatrix} \begin{matrix} 100 \\ 10000 \end{matrix}$$

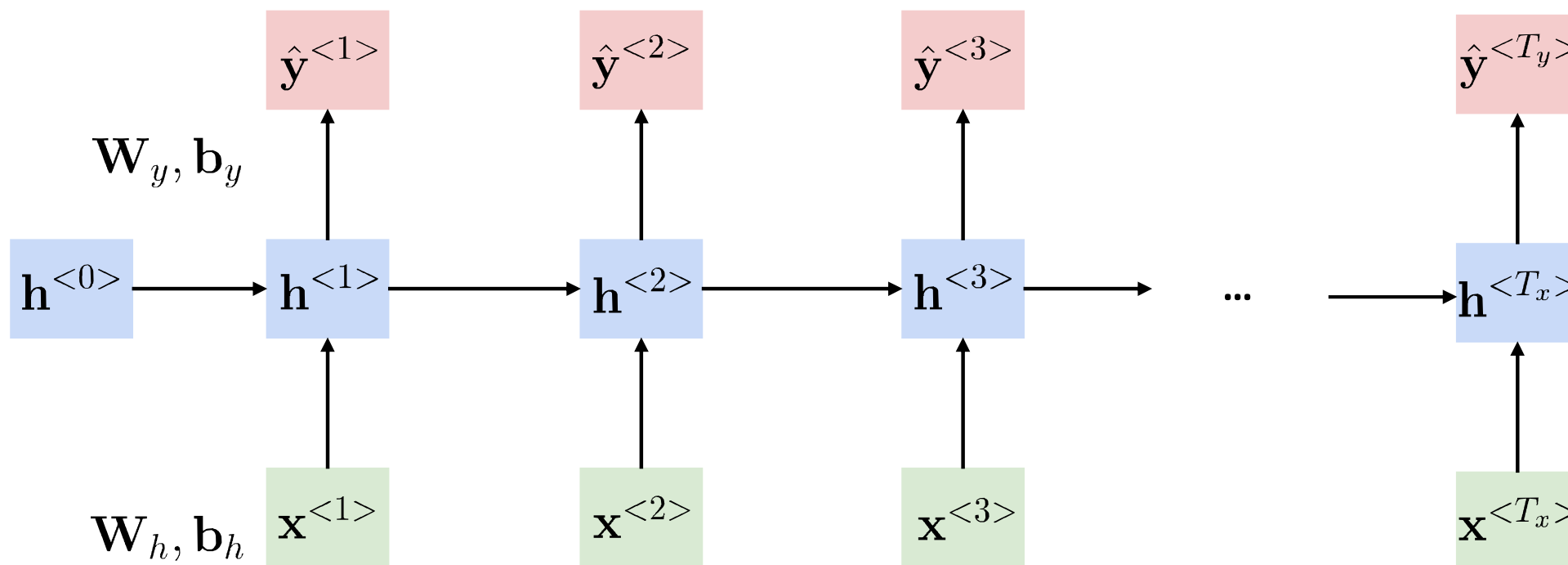
$$[\mathbf{W}_{hh}; \mathbf{W}_{hx}] \begin{bmatrix} \mathbf{h}^{<t-1>} \\ \mathbf{x}^{<t>} \end{bmatrix} = \mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>}$$



2

命名实体识别

→ 循环神经网络



$$\mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) = -y^{<t>} \log \hat{y}^{<t>} - (1 - y^{<t>}) \log(1 - \hat{y}^{<t>})$$

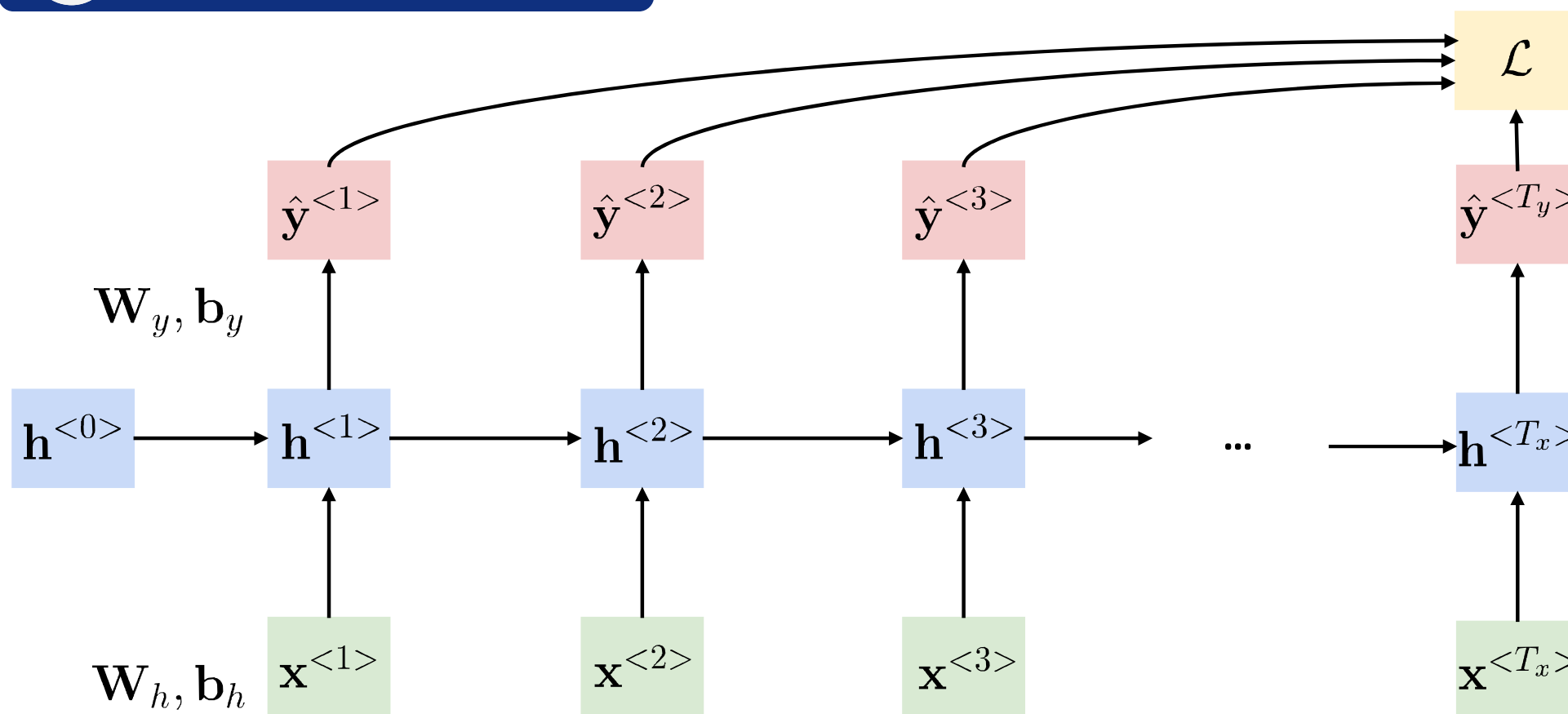
$$\mathcal{L}(\hat{y}, y) = \sum_{t=1}^{T_y} \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$



2

命名实体识别

→ 循环神经网络



$$\mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) = -y^{<t>} \log \hat{y}^{<t>} - (1 - y^{<t>}) \log(1 - \hat{y}^{<t>})$$

$$\mathcal{L}(\hat{y}, y) = \sum_{t=1}^{T_y} \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$



3 不同结构的RNN

	x	y
语音识别		→ “The quick brown fox jumped over the lazy dog”
音乐生成		→ 
情感分类	“There is nothing to like in this movie.”	→ 
机器翻译	“Voulez-vous chanter avec moi?”	→ “Do you want to sing with me?”
视频行为识别		→ Running
命名实体识别	“Yesterday, Harry Potter met Hermione Granger.”	→ “Yesterday, Harry Potter met Hermione Granger .”

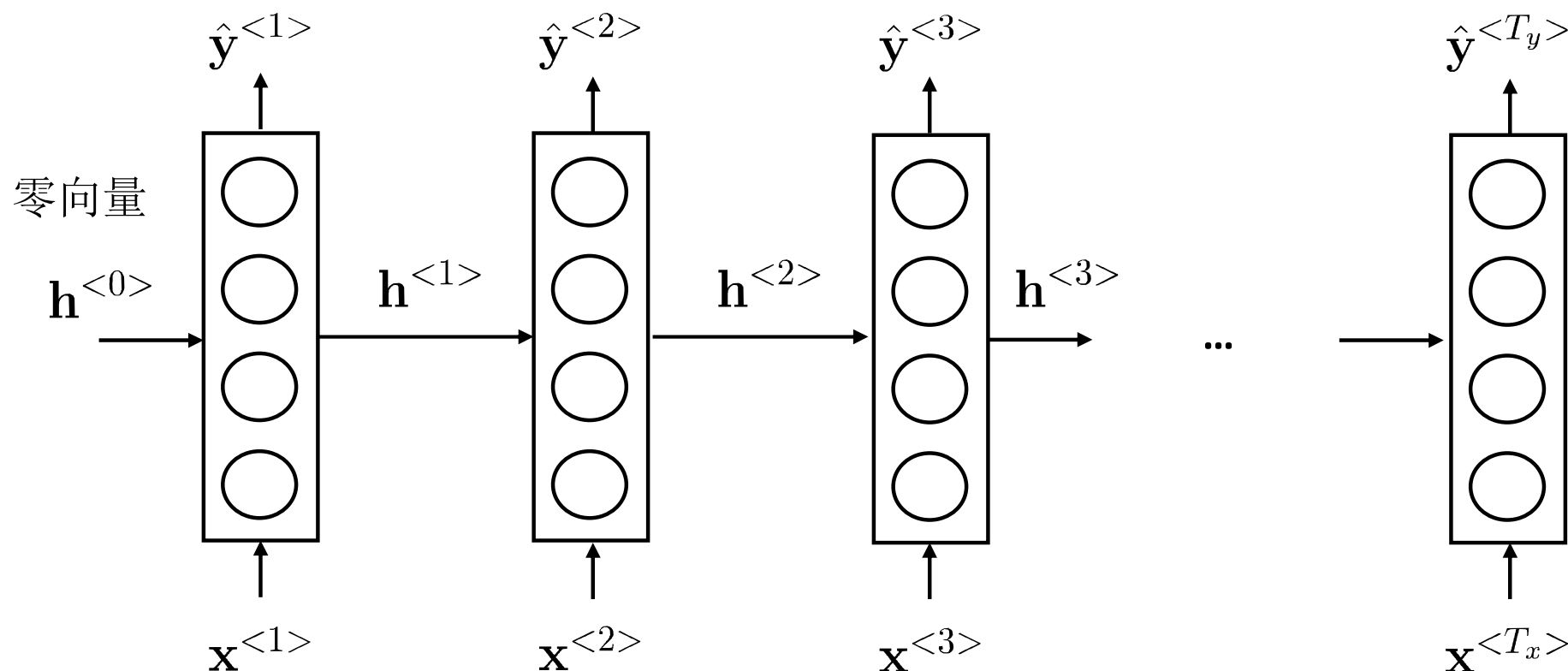
3 不同结构的RNN

■ Many-to-many (命名实体识别)

$$T_x = T_y$$

In fact, the **Chinese** **NORP** market has the **three** **CARDINAL** most influential names of the retail and tech space – **Alibaba** **ORG**, **Baidu** **ORG**, and **Tencent** **PERSON** (collectively touted as **BAT** **ORG**), and is betting big in the global **AI** **ORG** in retail industry space. The **three** **CARDINAL** giants which are claimed to have a cut-throat competition with the **U.S.** **ORG** (in terms of resources and capital) are positioning themselves to become the 'future **AI** **PERSON** platforms'. The trio is also expanding in other **Asian** **NORP** countries and investing heavily in the **U.S.** **ORG** based **AI** **ORG** startups to leverage the power of **AI** **ORG**. Backed by such powerful initiatives and presence of these conglomerates, the market in APAC AI is forecast to be the fastest-growing **one** **CARDINAL**, with an anticipated **CAGR** **PERSON** of **45%** **PERCENT** over **2018 - 2024** **DATE**.

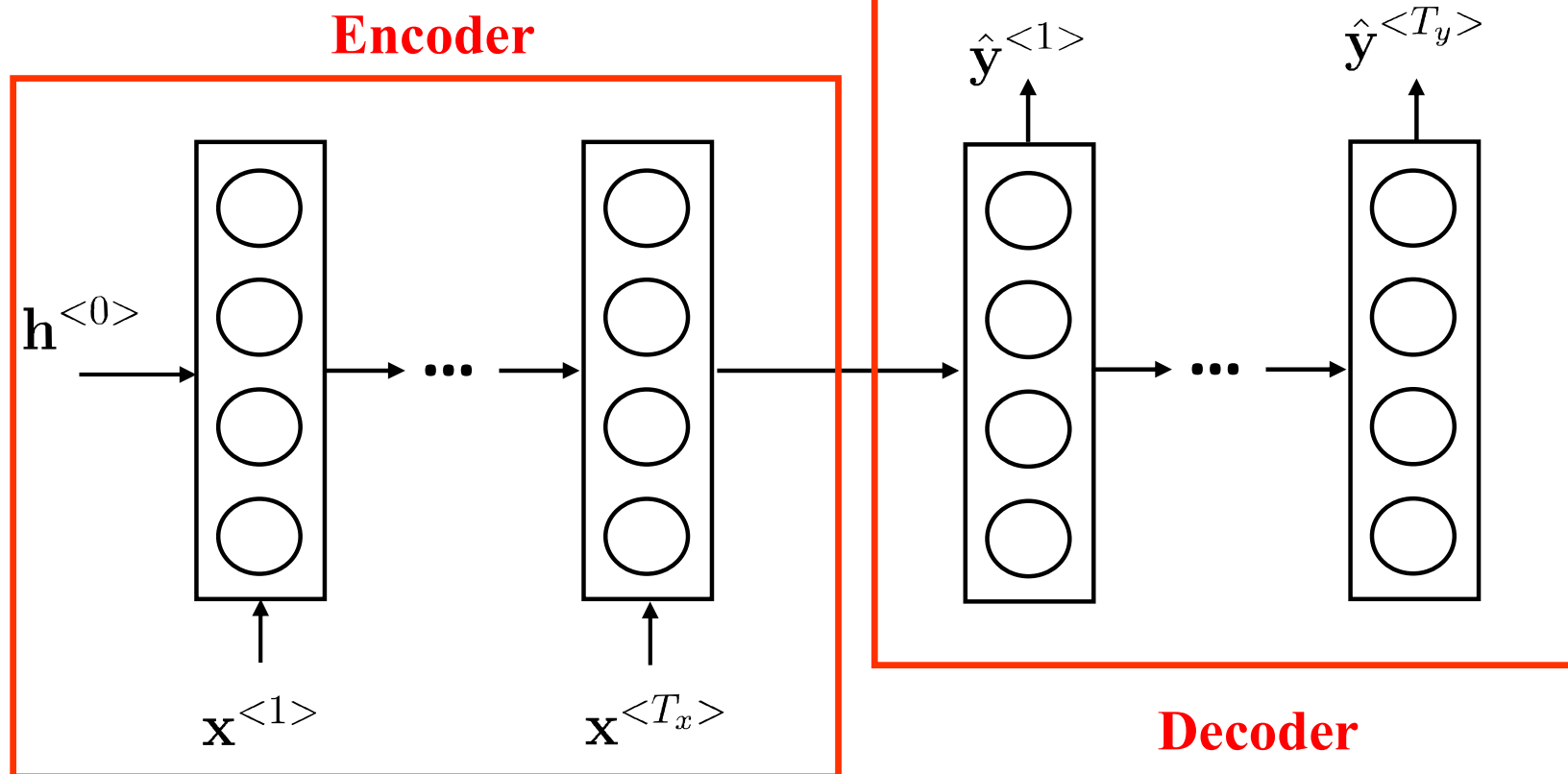
To further elaborate on the geographical trends, **North America** **LOC** has procured **more than 50%** **PERCENT** of the global share in **2017** **DATE** and has been leading the regional landscape of **AI** **ORG** in the retail market. The **U.S.** **ORG** has a significant credit in the regional trends with **over 65%** **PERCENT** of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as **Google** **ORG**, **IBM** **ORG**, and **Microsoft** **ORG**.



3 不同结构的RNN

■ Many-to-many (机器翻译)

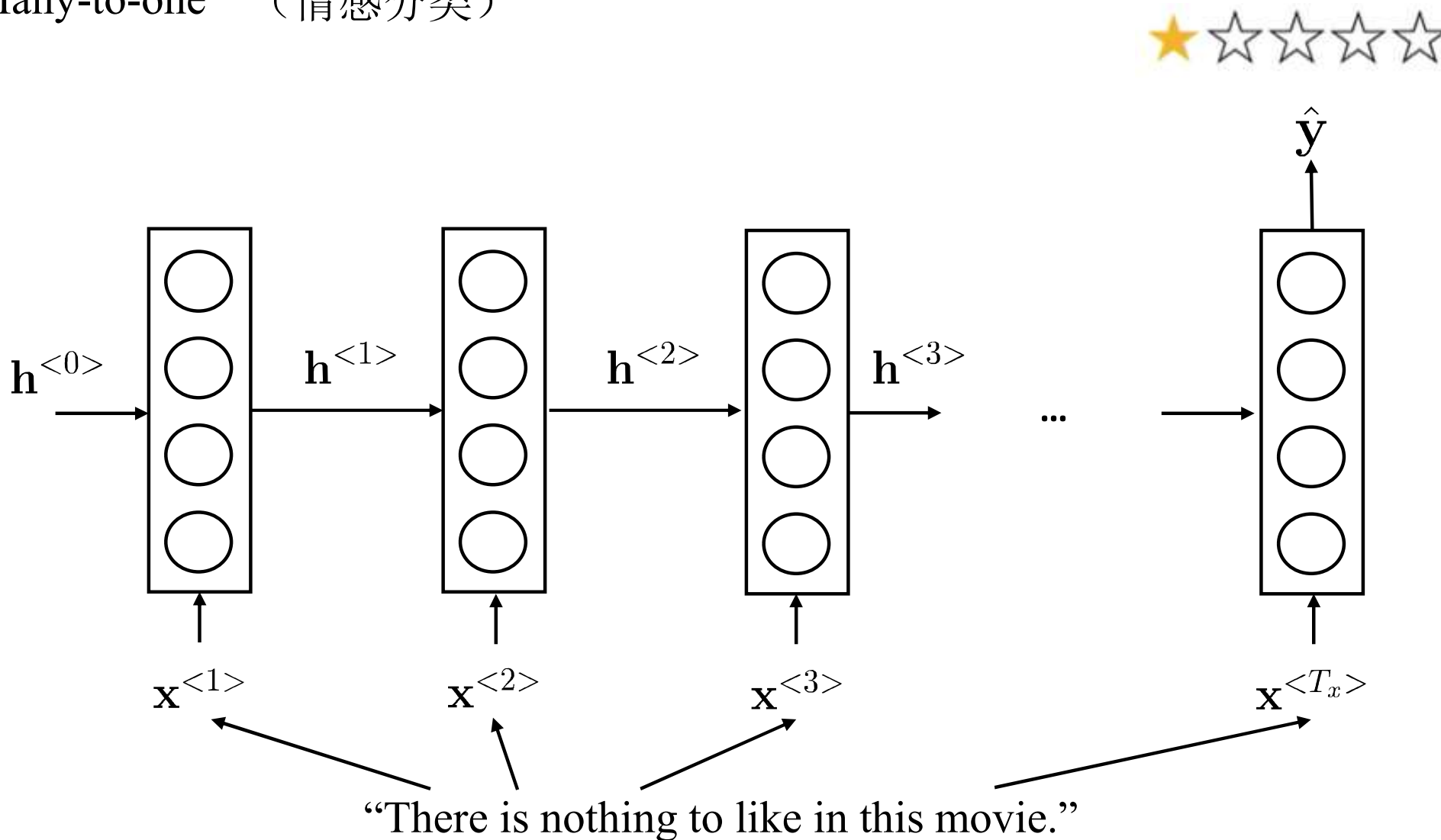
$$T_x \neq T_y$$



“Voulez-vous chanter avec moi?”

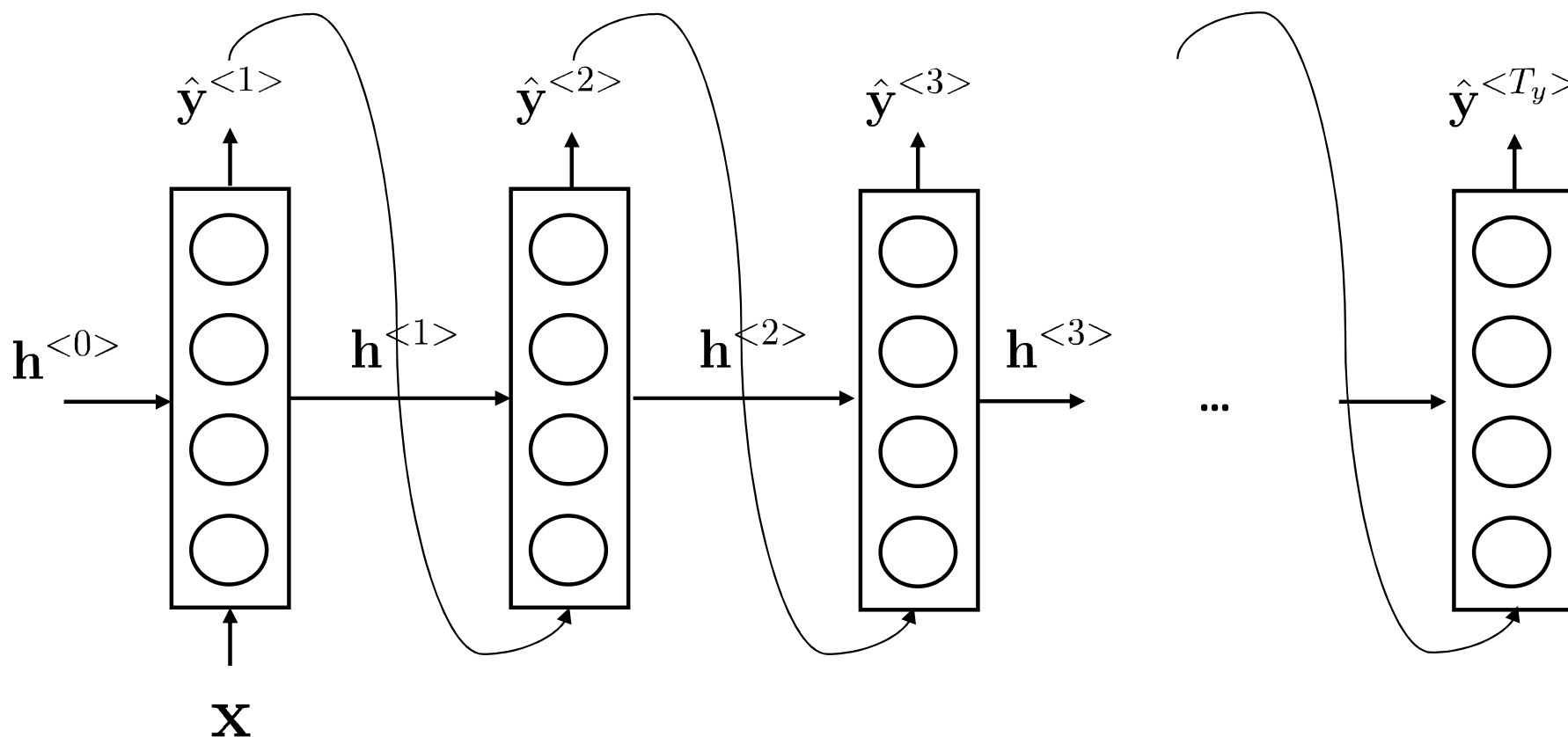
3 不同结构的RNN

■ Many-to-one (情感分类)



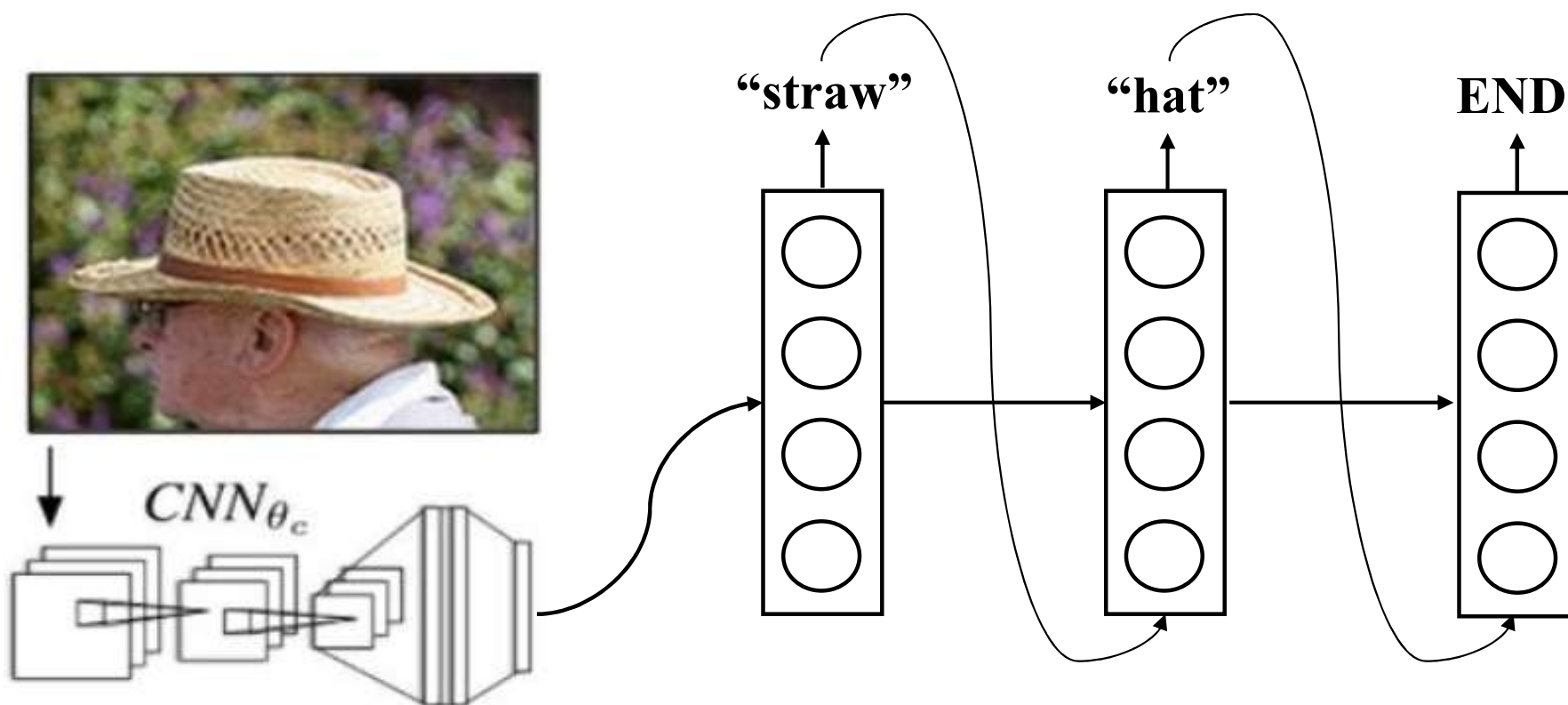
3 不同结构的RNN

■ One-to-many (音乐生成)



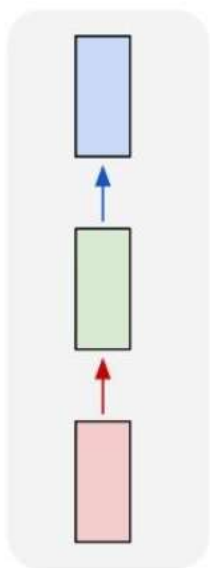
3 不同结构的RNN

■ One-to-many (Image caption)

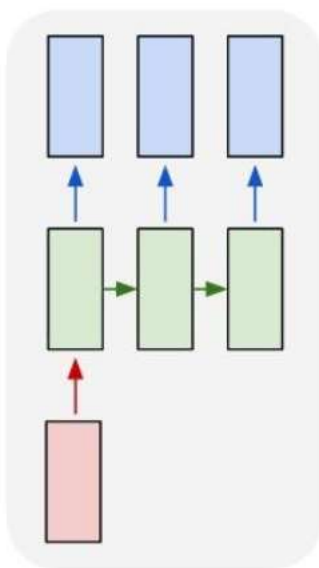


3 不同结构的RNN

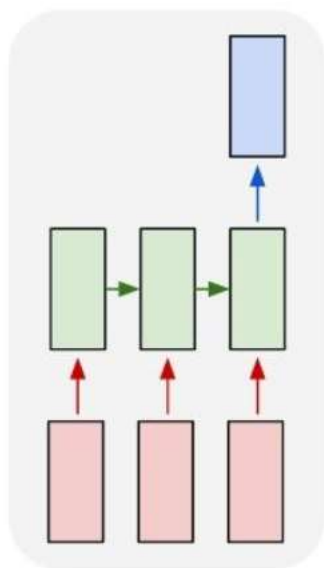
One-to-one



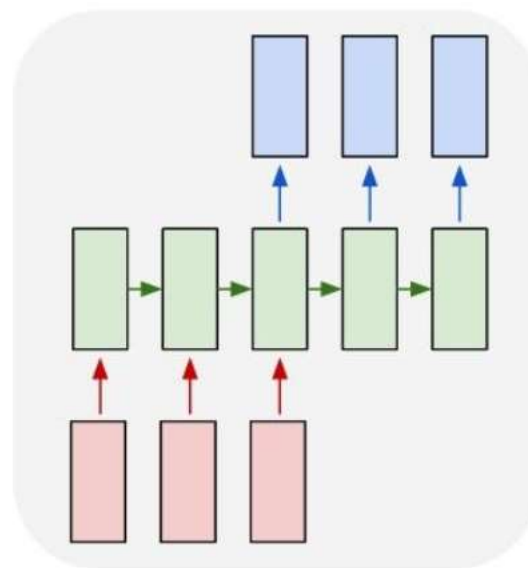
One-to-many



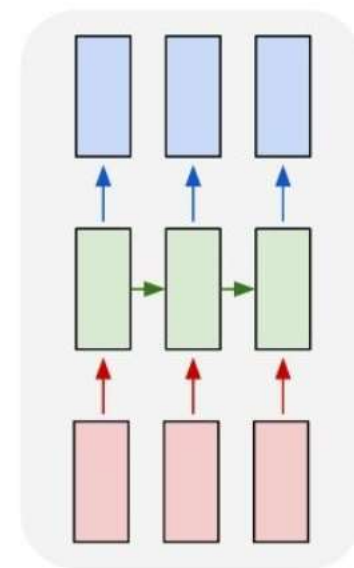
Many-to-one



Many-to-many



Many-to-many



- 语言模型 (Language model): 是自然语言处理的重要技术, 常用于语音识别和机器翻译等。
- 语言模型给一个序列 (句子) 分配概率, 找到最可能出现的词序列。同时估计某个序列中各单词出现的概率。

□ 语音识别系统: The apple and pair salad.

The apple and pear salad.

$$P(\text{The apple and pair salad})=3.2\text{e-}13$$

$$P(\text{The apple and pear salad})=5.7\text{e-}10$$

$$P(\text{Sentence})=? \quad \longrightarrow \quad P(\hat{y}^{<1>}, \hat{y}^{<2>}, \dots, \hat{y}^{<T_y>})$$



■ 构建语言模型

- 训练集：大型文本语料库
- 特征表示：one-hot向量

Cats average 15 hours of sleep a day.

$y^{<1>}$ $y^{<2>}$...

- 用RNN构建词序列的概率模型

Vocabulary

a
and
.
.
.
harry
.
.
.
potter
.
.
.
zulu

<EOS>

句尾标记

<UNK>

标点

<EOS>



Cats average 15 hours of sleep a day.

$$P(\mathbf{y}^{<1>}, \mathbf{y}^{<2>}, \mathbf{y}^{<3>}) = P(\mathbf{y}^{<1>})P(\mathbf{y}^{<2>}|\mathbf{y}^{<1>}) \cdot P(\mathbf{y}^{<3>}|\mathbf{y}^{<2>}, \mathbf{y}^{<1>})$$

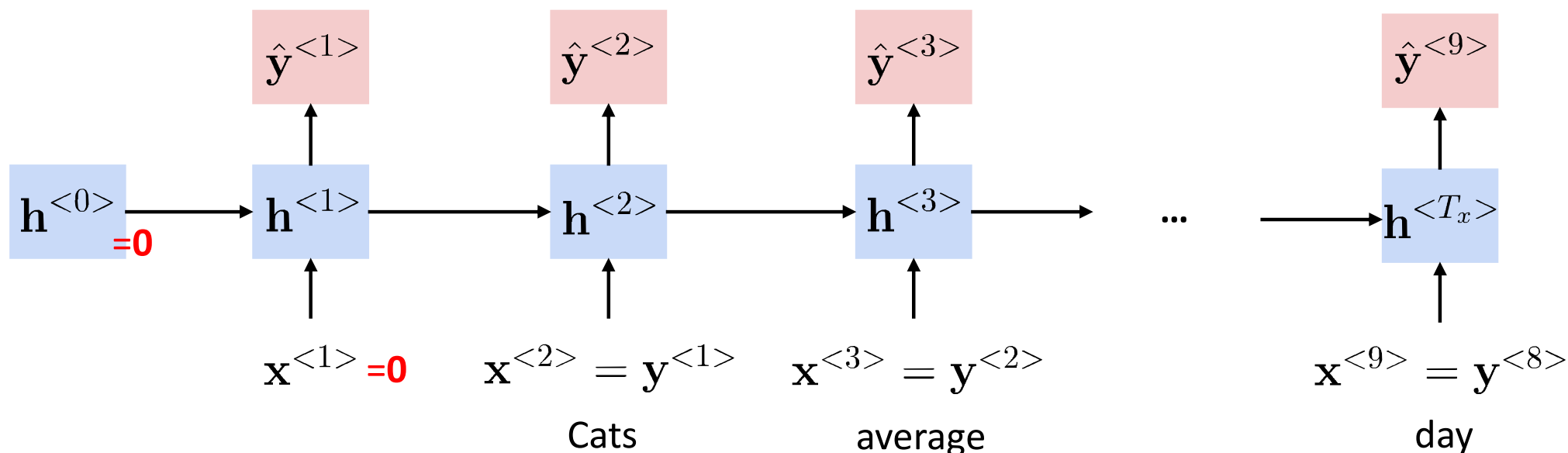
$$\rightarrow P(\mathbf{y}^{<1>}, \mathbf{y}^{<2>}, \dots, \mathbf{y}^{<T>}) = \prod_{t=1}^T P(\mathbf{y}^{<t>}|\mathbf{y}^{<1>}, \mathbf{y}^{<2>}, \dots, \mathbf{y}^{<t-1>})$$

$P(a), P(\text{and}), \dots, P(\text{cats}), \dots, P(\text{zulu}), P(<UNK>), P(<EOS>)$

$P(\text{cats})$

$P(\text{average}|\text{cats})$ $P(_|\text{"cats average"})$

$P(<EOS> | \text{"..."})$

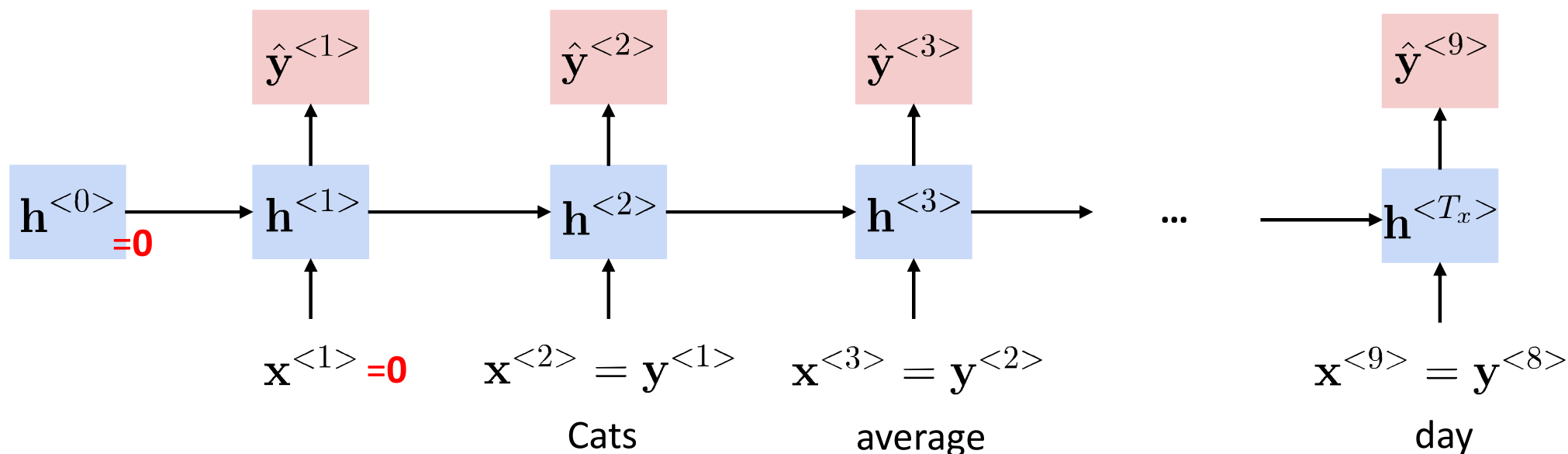


Cats average 15 hours of sleep a day.

$P(a), P(\text{and}), \dots, P(\text{cats}), \dots, P(\text{zulu}), P(< UNK >), P(< EOS >)$

$P(\text{cats}) \quad P(\text{average}|\text{cats}) \quad P(_ | \text{"cats average"})$

$P(< EOS > | \text{"..."})$

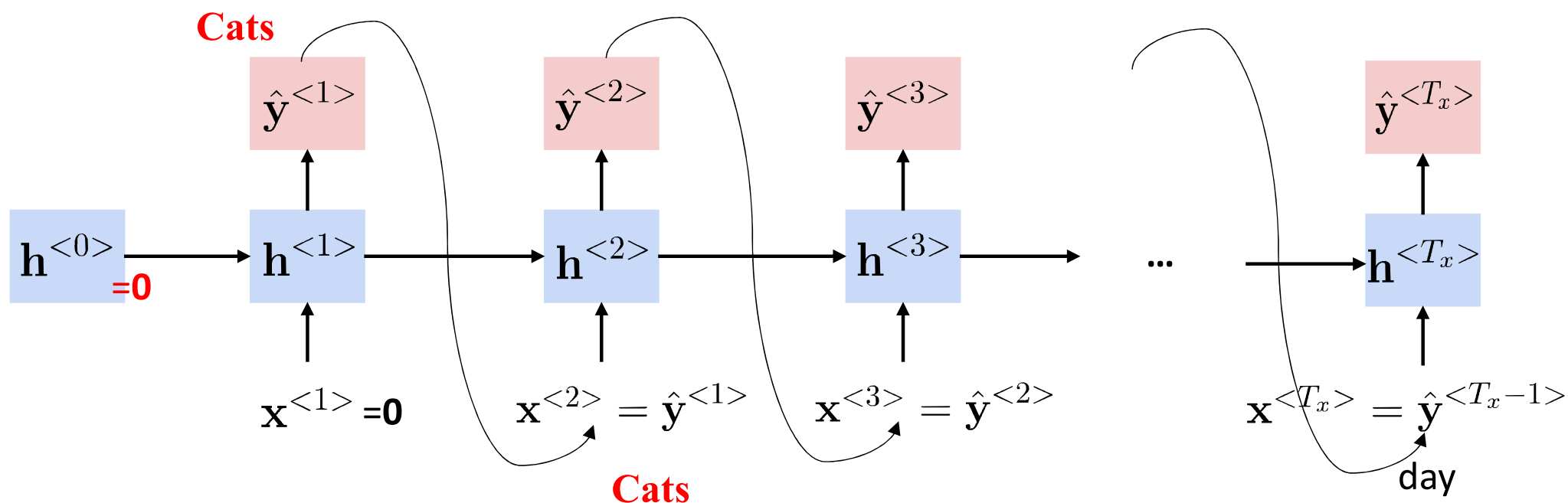


$$\mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) = - \sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

$$\mathcal{L}(\hat{y}, y) = \sum_{t=1}^{T_y} \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$

■ 用训练好的RNN生成句子

- 在每个时间步下，根据输出的概率分布进行采样。
- 将采样得到的单词作为下一个时间步的输入



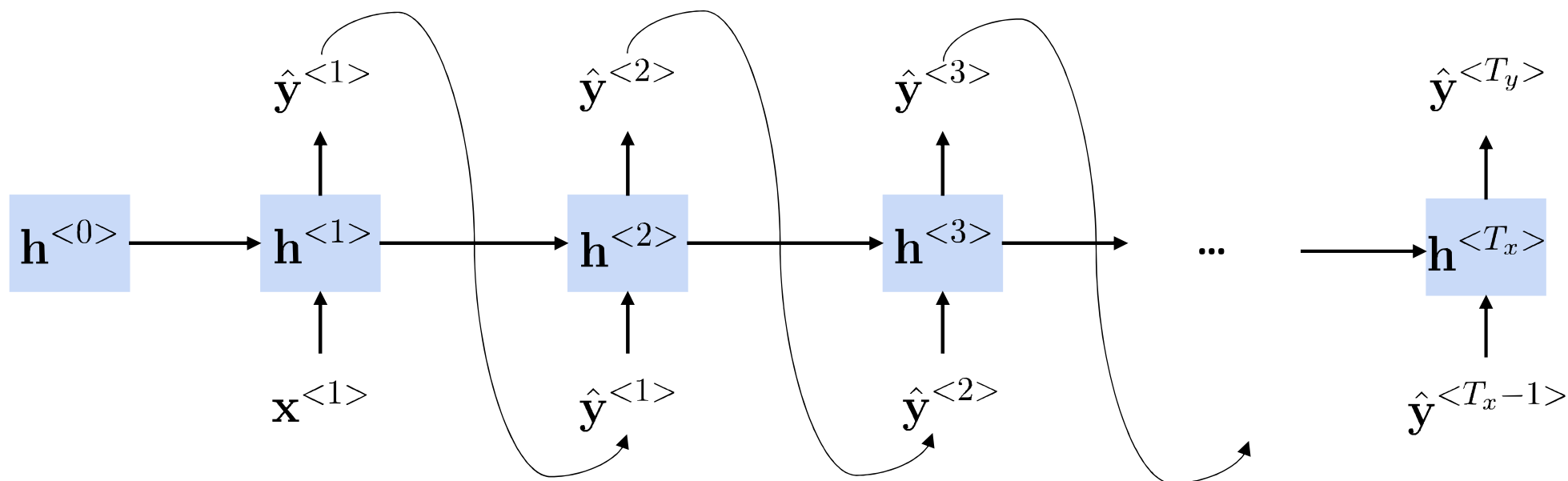
4

语言模型

■ 字符级别语言模型

Vocabulary=[a, and, ..., zulu, <UNK>]

Vocabulary=[a, b, c, ..., z, , ., ?, :, 0, 1, 2, ...]



■ 文本生成

News

President enrique peña nieto, announced
sench's sulk former coming football langston
paring.

"I was not at all surprised," said hich langston.

"Concussion epidemic", to be examined.

The gray football the told some and this has on
the uefa icon, should money as.

Shakespeare

The mortal moon hath her eclipse in love.

And subject of this thou art another this fold.

When besser be my love to me see sabl's.

For whose are ruse of mine eyes heaves.

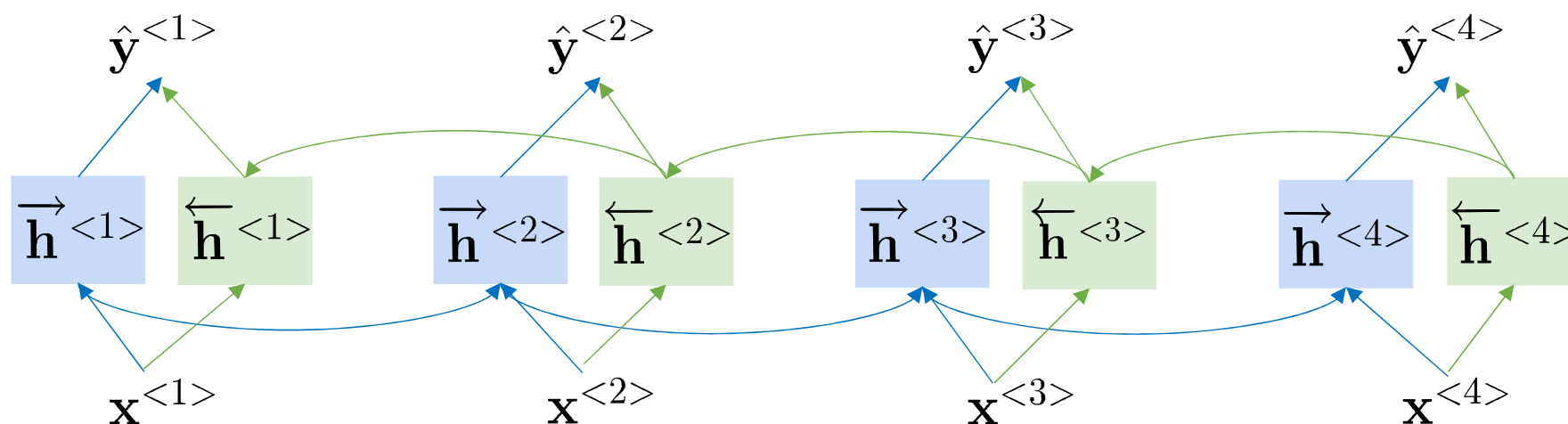


5 更多结构的RNN

- 双向循环神经网络 (Bi-directional RNN)
 - 获取历史数据的信息 + 获取未来数据的信息

He said, “Teddy bears are on sale!”

He said, “Teddy Roosevelt was a great President!”



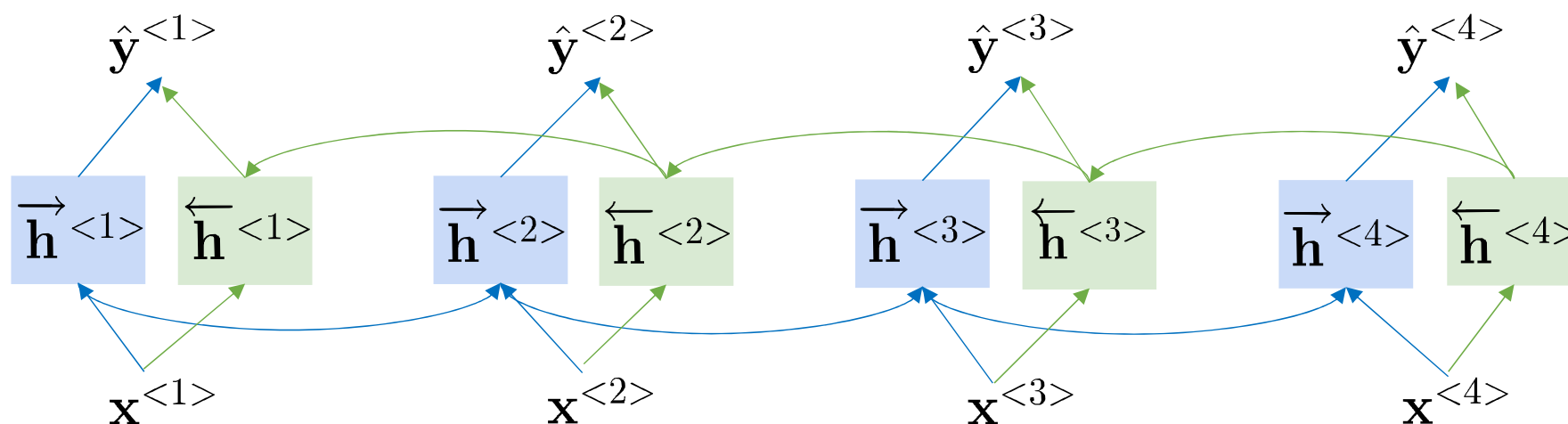
$$\hat{y}^{<t>} = g(\mathbf{W}_y[\vec{\mathbf{h}}^{<t>}, \overleftarrow{\mathbf{h}}^{<t>}] + \mathbf{b}_y)$$

5 更多结构的RNN

■ 双向循环神经网络 (Bi-directional RNN)

□ 获取历史数据的信息 + 获取未来数据的信息

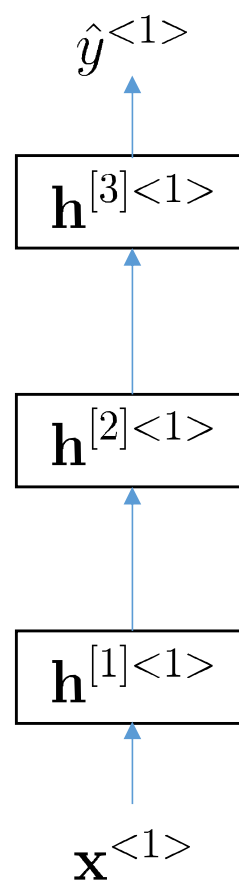
缺陷 ◆ 需要获取完整的序列（句子）才能进行预测，不能进行实时处理



5 更多结构的RNN

■ 深层循环神经网络 (Deep RNN)

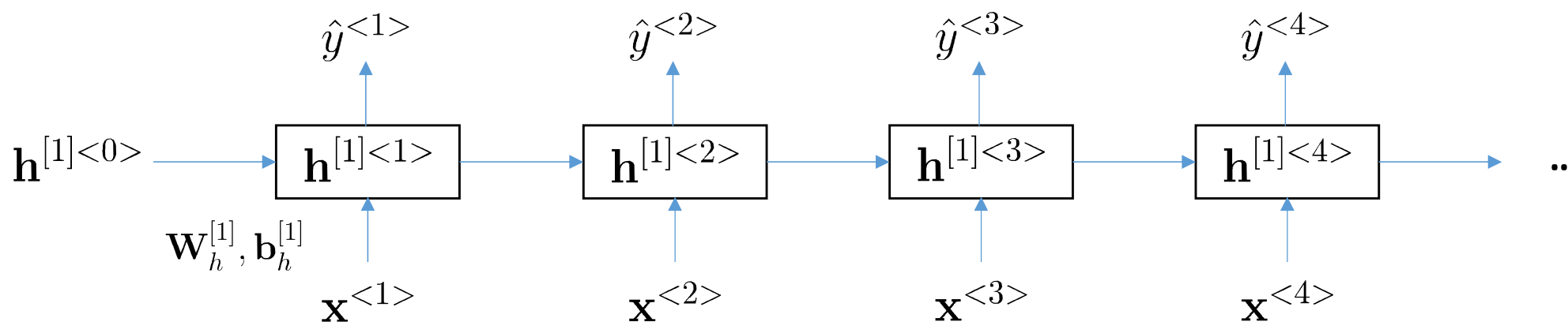
前馈神经网络



5 更多结构的RNN

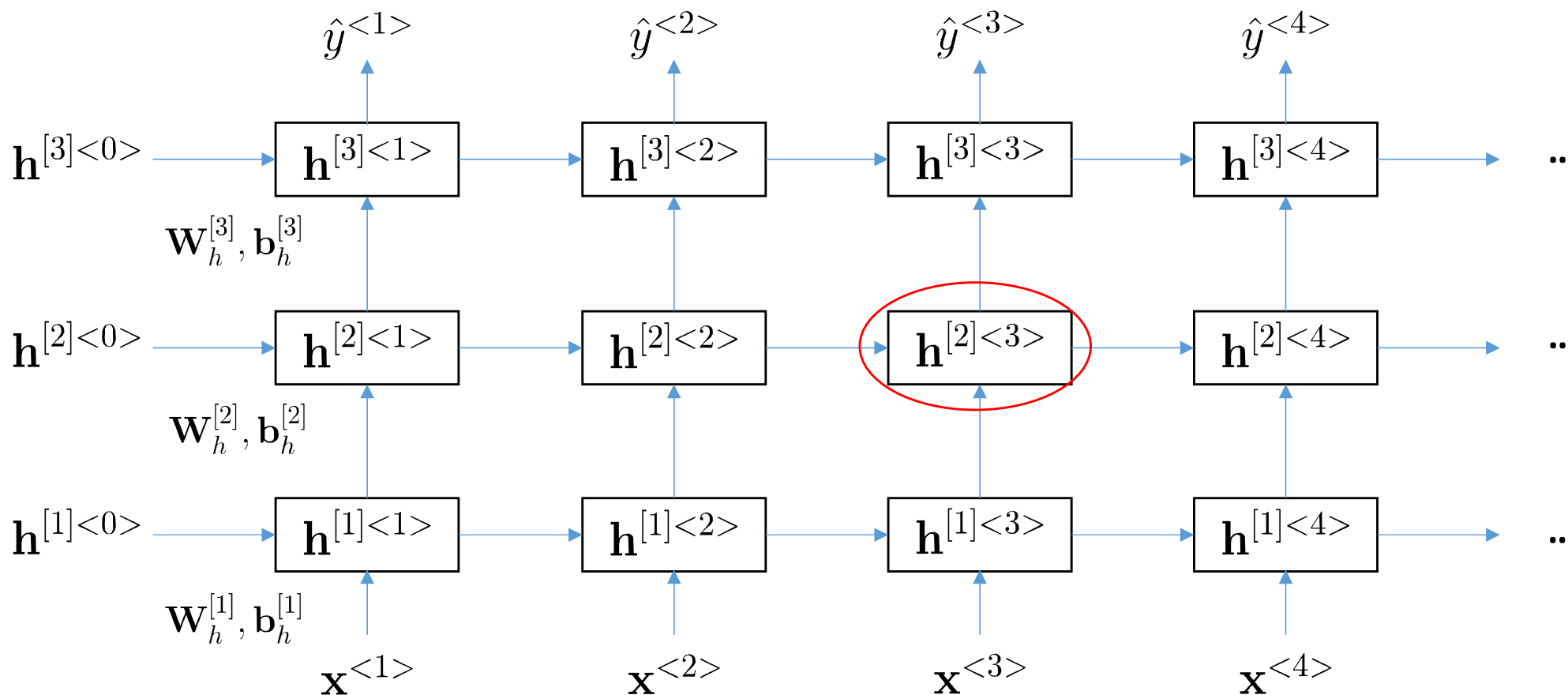
■ 深层循环神经网络 (Deep RNN)

循环神经网络



5 更多结构的RNN

■ 深层循环神经网络 (Deep RNN)



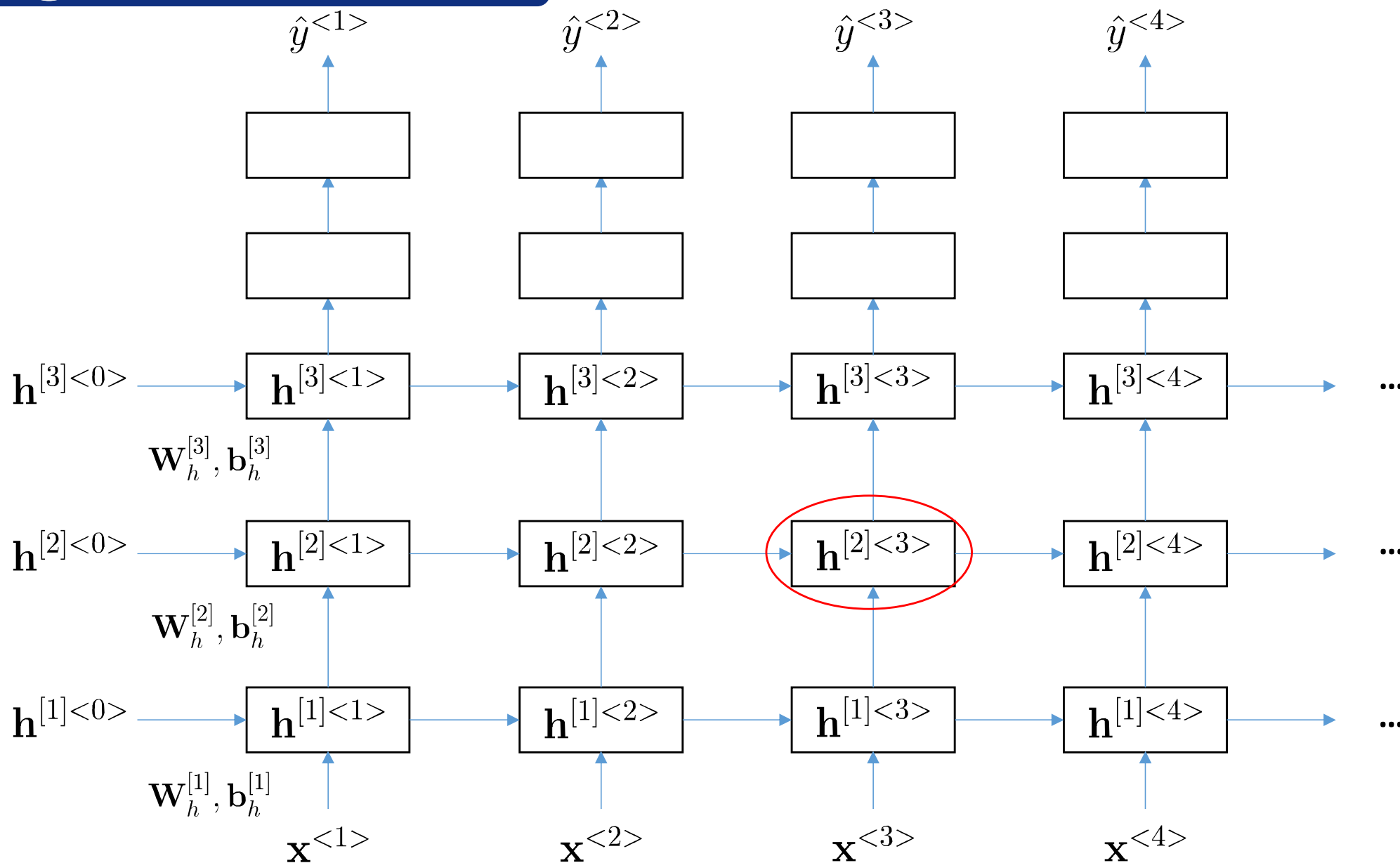
$$\mathbf{h}^{[2]<3>} = g(\mathbf{W}_h^{[2]}[\mathbf{h}^{[2]<2>}, \mathbf{h}^{[1]<3>}] + \mathbf{b}_h^{[2]})$$

$$\mathbf{h}^{[l]<t>} = g(\mathbf{W}_h^{[l]}[\mathbf{h}^{[l]<t-1>}, \mathbf{h}^{[l-1]<t>}] + \mathbf{b}_h^{[l]})$$



4 深层循环神经网络

Deep RNN



03

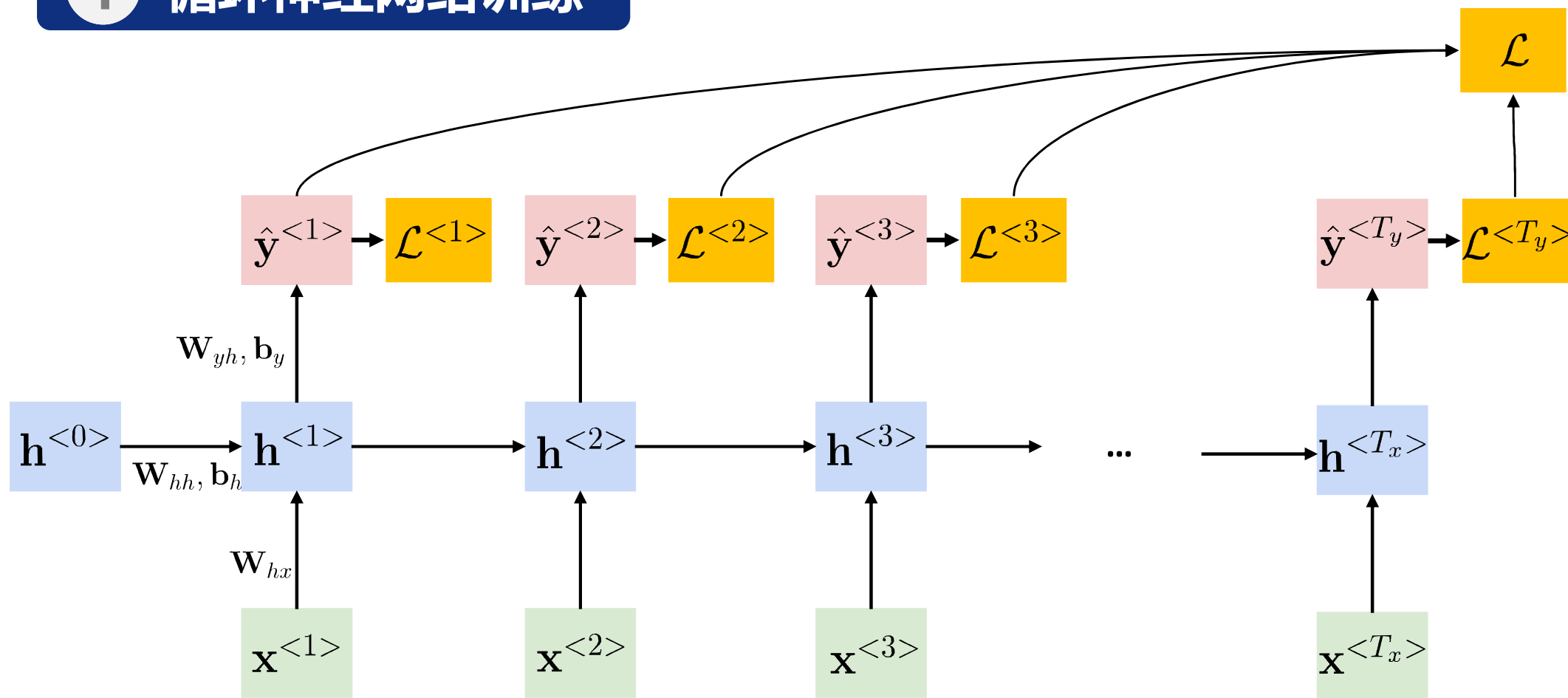
PART THREE

循环神经网络训练

RNN Training



1 循环神经网络训练



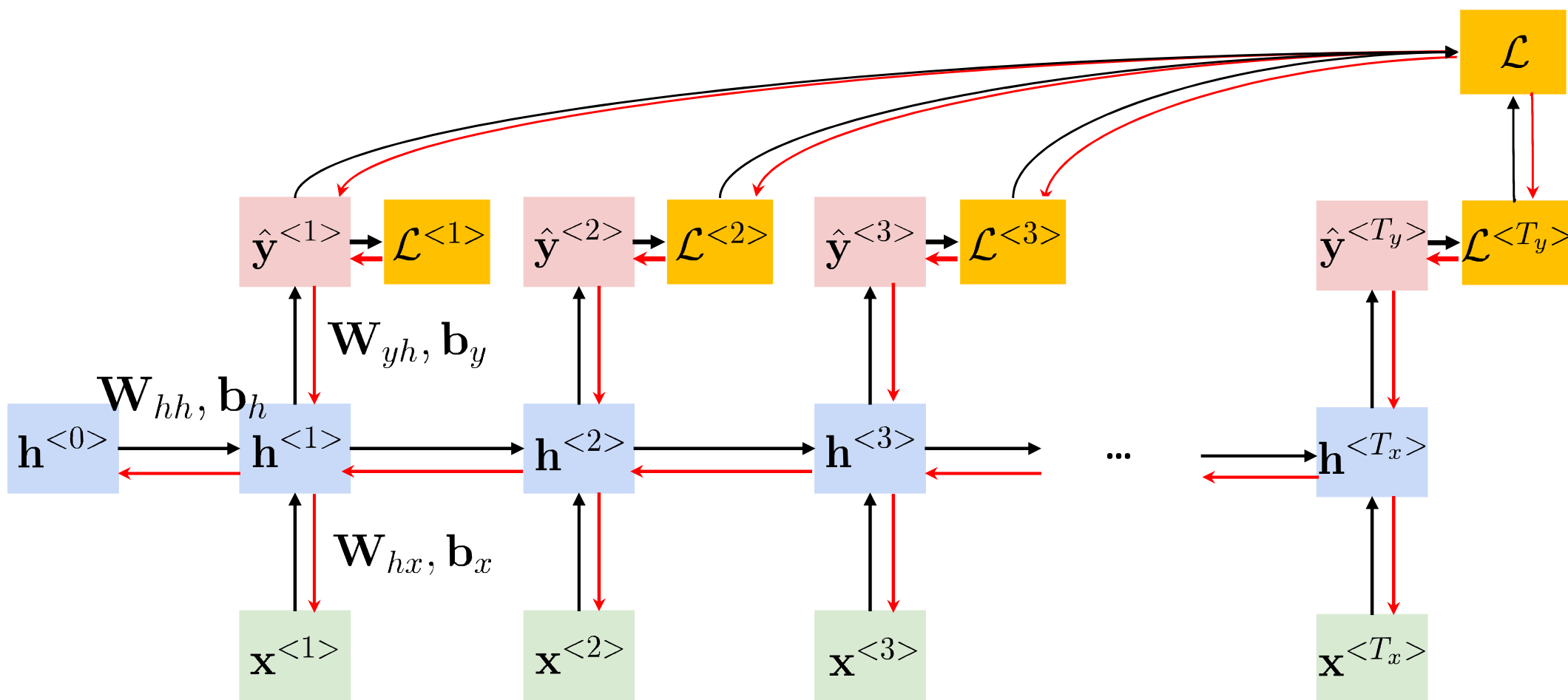
$$\mathcal{L}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>}) = - \sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

$$\mathcal{L} = \sum_t \mathcal{L}^{<t>}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>})$$



1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播

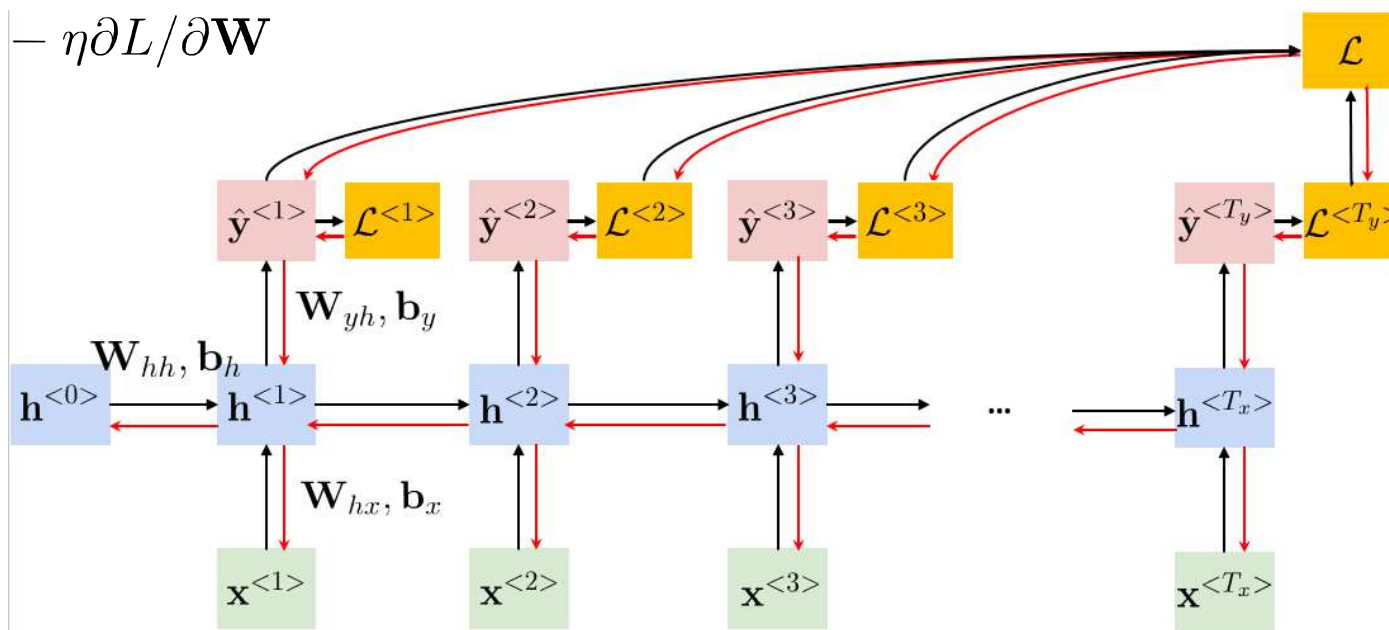


1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播

BPTT的中心思想和BP算法相同，本质是梯度下降法。需要寻优的参数有三个， $\mathbf{W}_{hh}$, $\mathbf{W}_{hx}$, $\mathbf{W}_{yh}$. 与BP算法不同的是， $\mathbf{W}_{hh}$, $\mathbf{W}_{hx}$ 参数在寻优的过程中需要追溯之前历史数据，因此RNN有记忆功能。

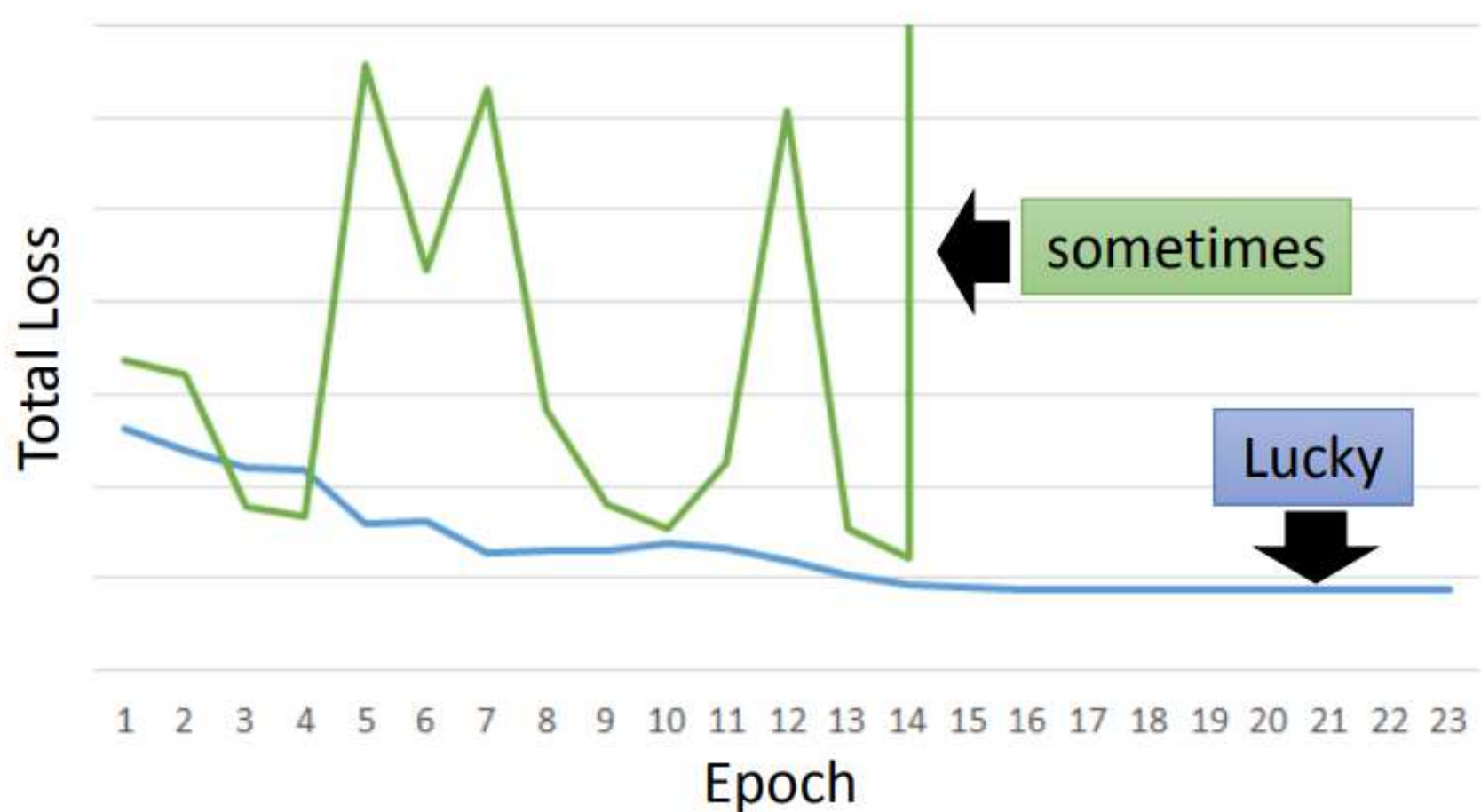
$$\mathbf{W} \leftarrow \mathbf{W} - \eta \partial L / \partial \mathbf{W}$$



1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播

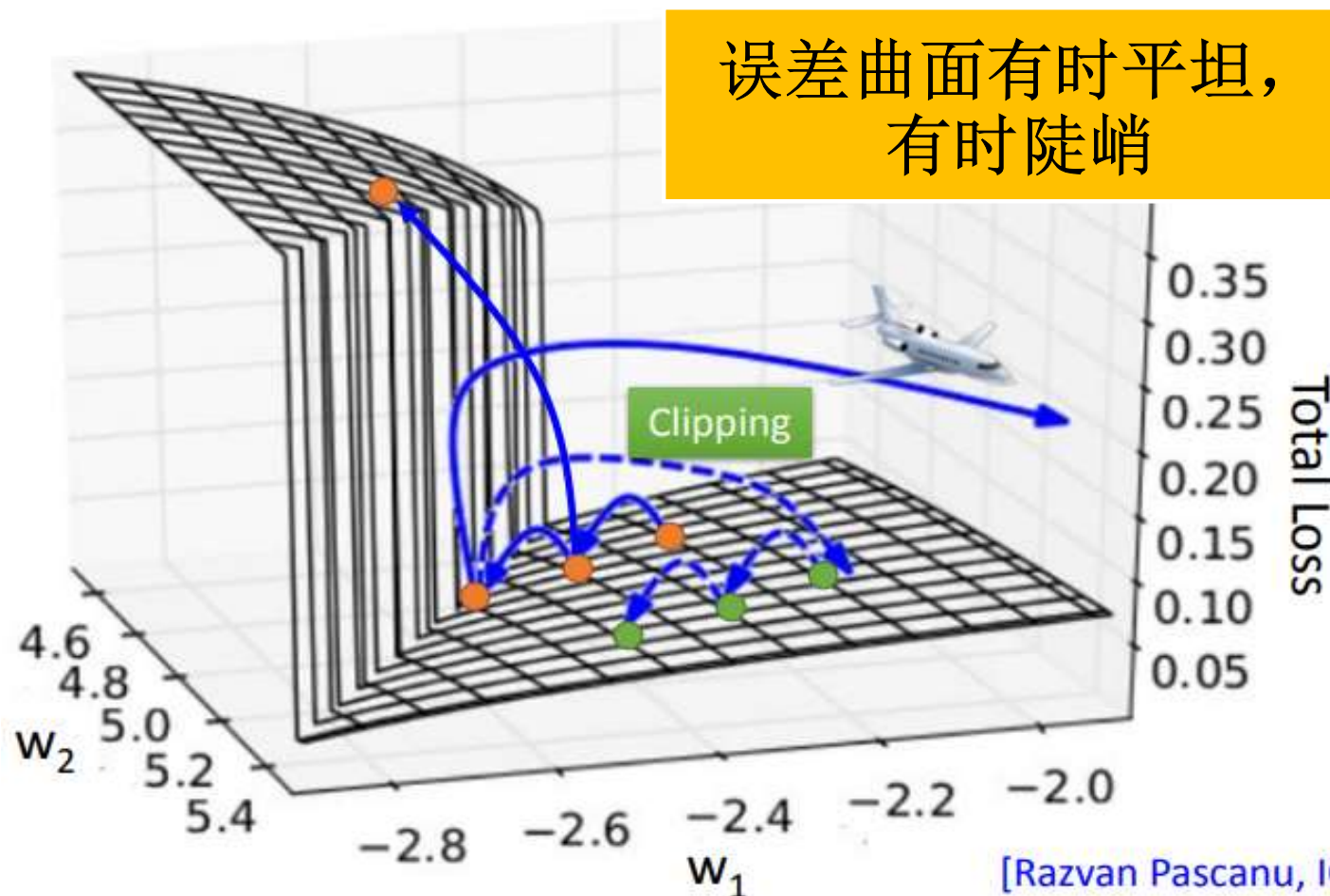
□ 训练RNN通常比较困难



1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播

□ 训练RNN通常比较困难

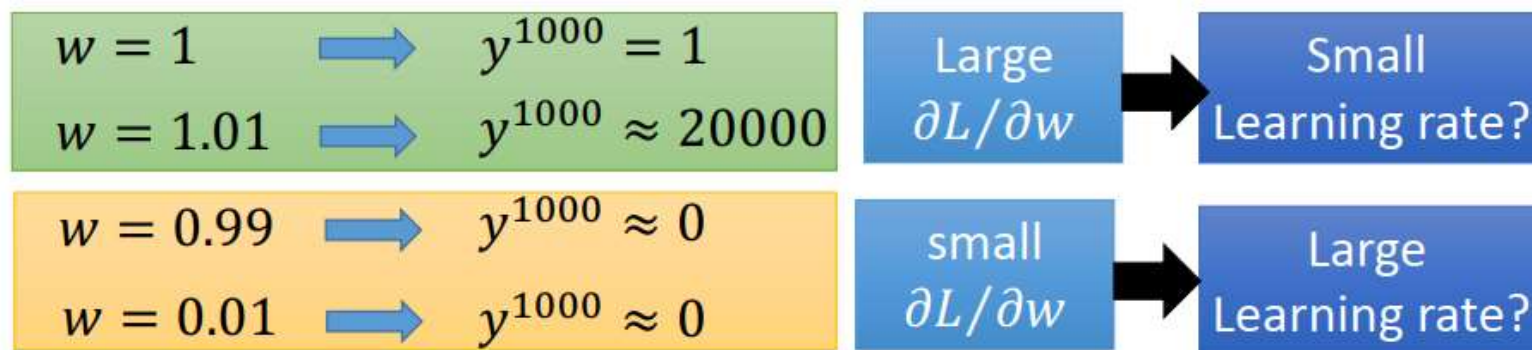


[Razvan Pascanu, ICML'13]

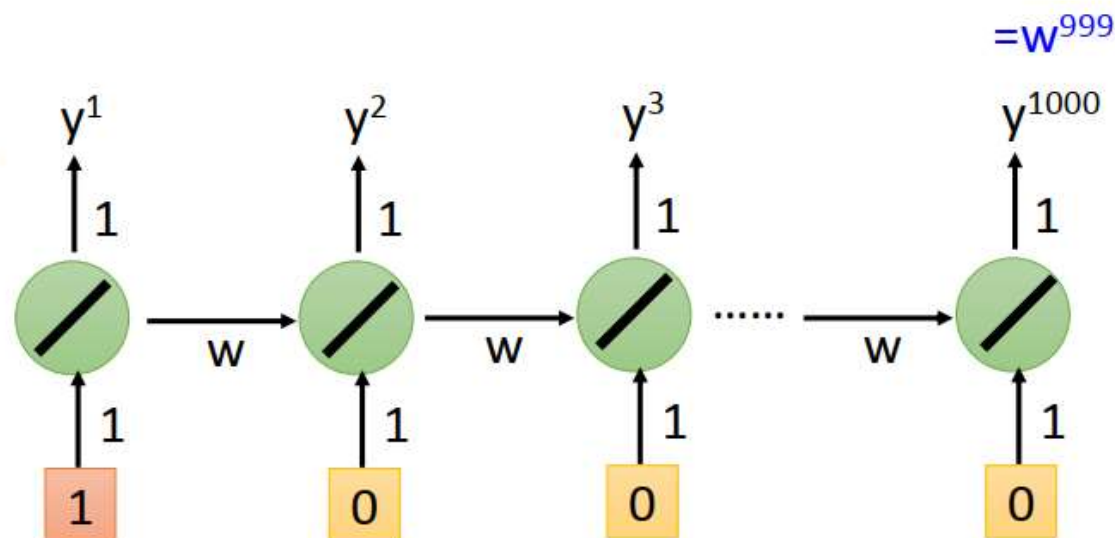
1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播

□ 训练RNN通常比较困难



Toy Example



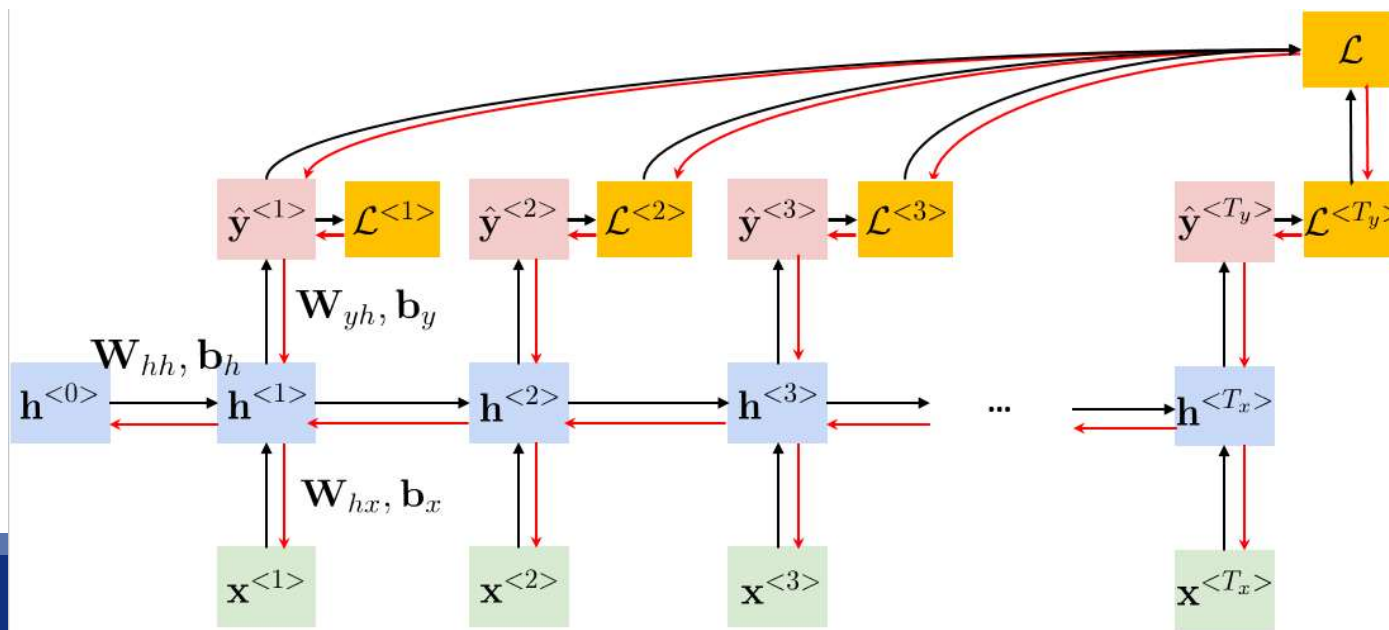
1 循环神经网络训练

- BPTT (Back-propagation through time): 随时间反向传播
- 对 $\mathbf{W}_{yh}$ 求偏导:

$$\frac{\partial \mathcal{L}^{<t>}}{\partial \mathbf{W}_{yh}} = \frac{\partial \mathcal{L}^{<t>}}{\partial \hat{y}^{<t>}} \cdot \frac{\partial \hat{y}^{<t>}}{\partial \mathbf{W}_{yh}}$$

其中嵌套着激活函数函数，
是复合函数的求导过程。

$$\begin{aligned} \mathbf{h}^{<t>} &= g(\mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>} + \mathbf{b}_h) \\ \hat{y}^{<t>} &= g_2(\mathbf{W}_{yh}\mathbf{h}^{<t>} + \mathbf{b}_y) \\ \mathcal{L}(\hat{y}^{<t>}, y^{<t>}) &= -\sum_i y_i^{<t>} \log \hat{y}_i^{<t>} \\ \mathcal{L} &= \sum_t \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>}) \end{aligned}$$



1 循环神经网络训练

- 对 $\mathbf{W}_{hh}$ 求偏导:

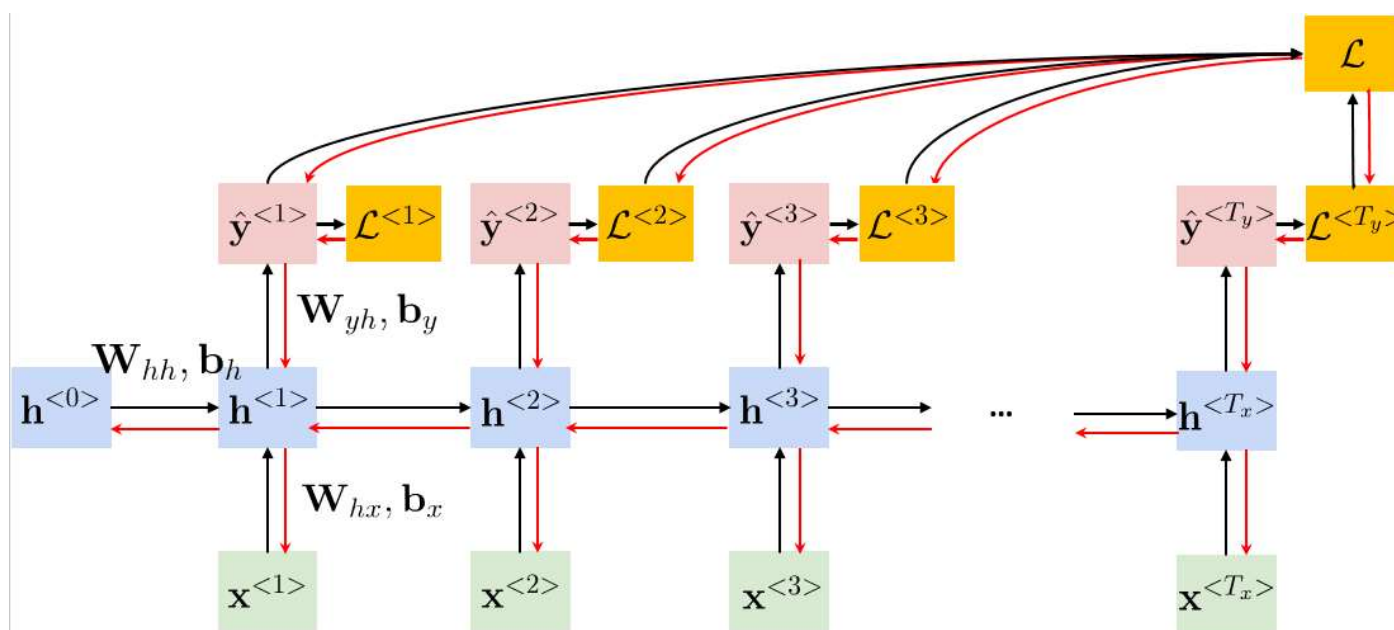
$$\begin{aligned} \frac{\partial \mathcal{L}^{<3>}}{\partial \mathbf{W}_{hh}} &= \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{W}_{hh}} \\ &+ \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{h}^{<2>}} \cdot \frac{\partial \mathbf{h}^{<2>}}{\partial \mathbf{W}_{hh}} \\ &+ \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{h}^{<2>}} \cdot \frac{\partial \mathbf{h}^{<2>}}{\partial \mathbf{h}^{<1>}} \cdot \frac{\partial \mathbf{h}^{<1>}}{\partial \mathbf{W}_{hh}} \end{aligned}$$

$$\mathbf{h}^{<t>} = g(\mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>} + \mathbf{b}_h)$$

$$\hat{y}^{<t>} = g_2(\mathbf{W}_{yh}\mathbf{h}^{<t>} + \mathbf{b}_y)$$

$$\mathcal{L}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>}) = -\sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

$$\mathcal{L} = \sum_t \mathcal{L}^{<t>}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>})$$



1 循环神经网络训练

- 对 $\mathbf{W}_{hx}$ 求偏导:

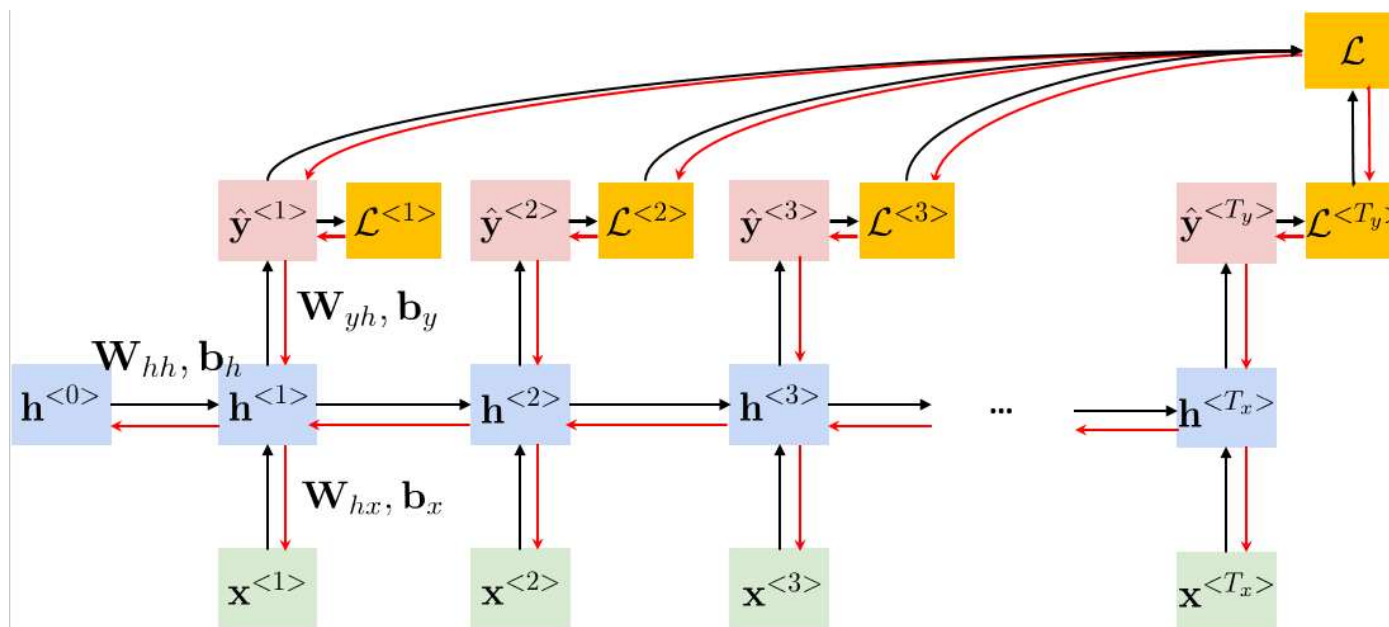
$$\begin{aligned} \frac{\partial \mathcal{L}^{<3>}}{\partial \mathbf{W}_{hx}} &= \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{W}_{hx}} \\ &+ \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{h}^{<2>}} \cdot \frac{\partial \mathbf{h}^{<2>}}{\partial \mathbf{W}_{hx}} \\ &+ \frac{\partial \mathcal{L}^{<3>}}{\partial \hat{y}^{<3>}} \cdot \frac{\partial \hat{y}^{<3>}}{\partial \mathbf{h}^{<3>}} \cdot \frac{\partial \mathbf{h}^{<3>}}{\partial \mathbf{h}^{<2>}} \cdot \frac{\partial \mathbf{h}^{<2>}}{\partial \mathbf{h}^{<1>}} \cdot \frac{\partial \mathbf{h}^{<1>}}{\partial \mathbf{W}_{hx}} \end{aligned}$$

$$\mathbf{h}^{<t>} = g(\mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>} + \mathbf{b}_h)$$

$$\hat{y}^{<t>} = g_2(\mathbf{W}_{yh}\mathbf{h}^{<t>} + \mathbf{b}_y)$$

$$\mathcal{L}(\hat{y}^{<t>}, y^{<t>}) = - \sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

$$\mathcal{L} = \sum_t \mathcal{L}^{<t>}(\hat{y}^{<t>}, y^{<t>})$$



1 循环神经网络训练

- 对 $\mathbf{W}_{hh}$ 求偏导:

$$\frac{\partial \mathcal{L}^{<t>}}{\partial \mathbf{W}_{hh}} = \sum_{k=0}^t \frac{\partial \mathcal{L}^{<t>}}{\partial \hat{y}^{<t>}} \cdot \frac{\hat{y}^{<t>}}{\partial \mathbf{h}^{<t>}} \cdot \left(\prod_{j=k+1}^t \frac{\partial \mathbf{h}^{<j>}}{\partial \mathbf{h}^{<j-1>}} \right) \cdot \frac{\partial \mathbf{h}^{<k>}}{\partial \mathbf{W}_{hh}}$$

$$\prod_{j=k+1}^t (\tanh)' \cdot \mathbf{W}_{hh}$$

- 对 $\mathbf{W}_{hx}$ 求偏导:

$$\frac{\partial \mathcal{L}^{<t>}}{\partial \mathbf{W}_{hx}} = \sum_{k=0}^t \frac{\partial \mathcal{L}^{<t>}}{\partial \hat{y}^{<t>}} \cdot \frac{\hat{y}^{<t>}}{\partial \mathbf{h}^{<t>}} \cdot \left(\prod_{j=k+1}^t \frac{\partial \mathbf{h}^{<j>}}{\partial \mathbf{h}^{<j-1>}} \right) \cdot \frac{\partial \mathbf{h}^{<k>}}{\partial \mathbf{W}_{hx}}$$

$$\prod_{j=k+1}^t (\text{sigmoid})' \cdot \mathbf{W}_{hh}$$

连乘会导致激活函数导数的连乘，进而会导致“梯度消失”和“梯度爆炸”现象的发生

$$\mathbf{h}^{<t>} = g(\mathbf{W}_{hh}\mathbf{h}^{<t-1>} + \mathbf{W}_{hx}\mathbf{x}^{<t>} + \mathbf{b}_h)$$

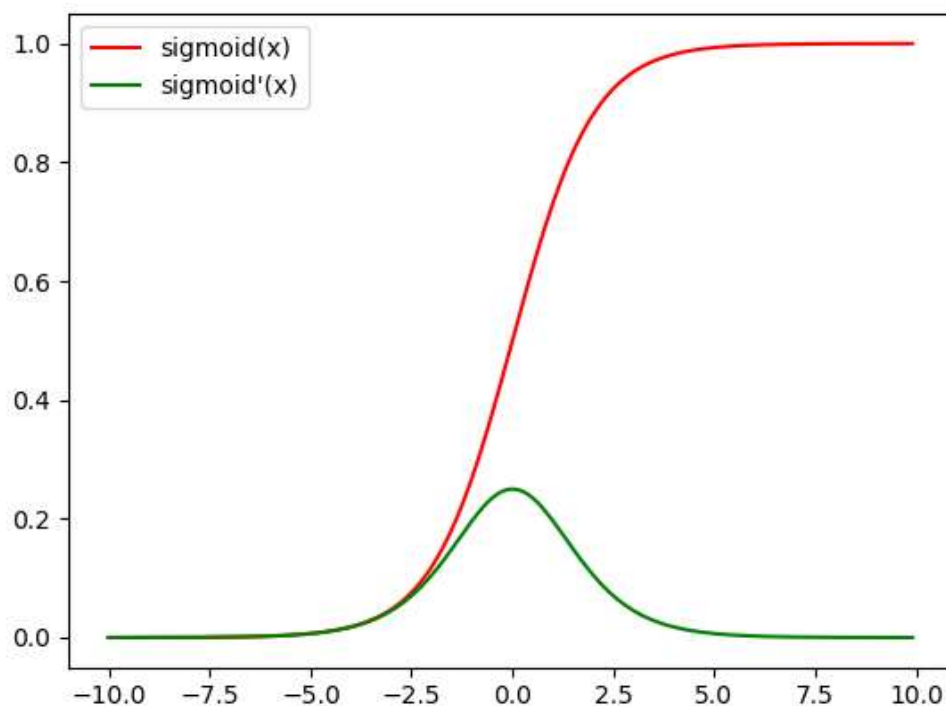
$$\hat{y}^{<t>} = g_2(\mathbf{W}_{yh}\mathbf{h}^{<t>} + \mathbf{b}_y)$$

$$\mathcal{L}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>}) = - \sum_i y_i^{<t>} \log \hat{y}_i^{<t>}$$

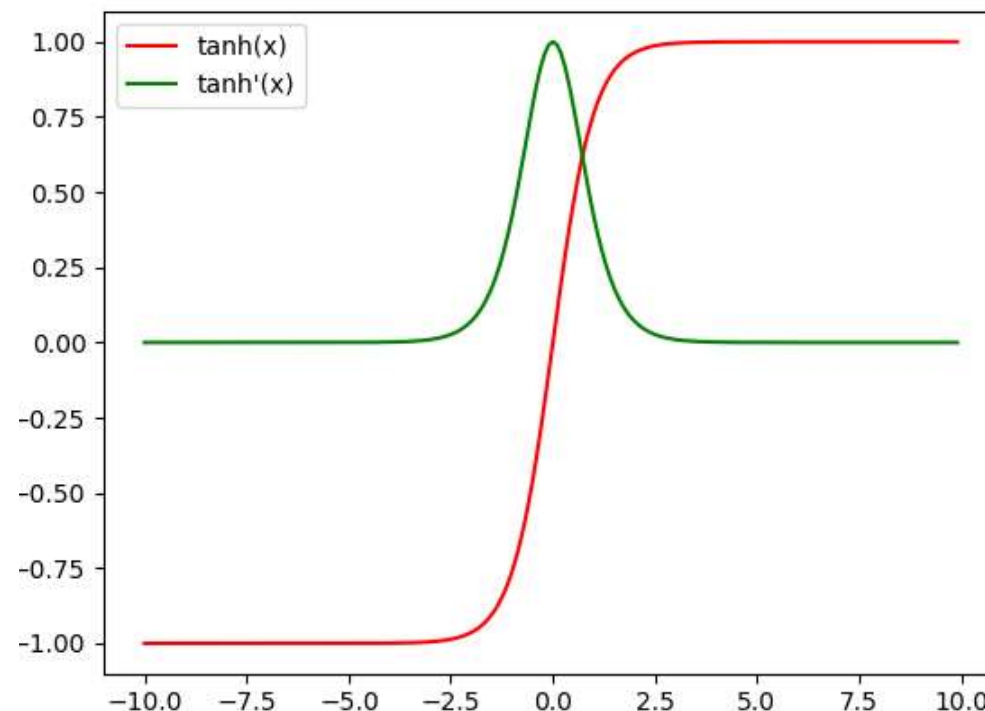
$$\mathcal{L} = \sum_t \mathcal{L}^{<t>}(\hat{\mathbf{y}}^{<t>}, \mathbf{y}^{<t>})$$



1 循环神经网络训练



Sigmoid函数及其导数



Tanh函数及其导数

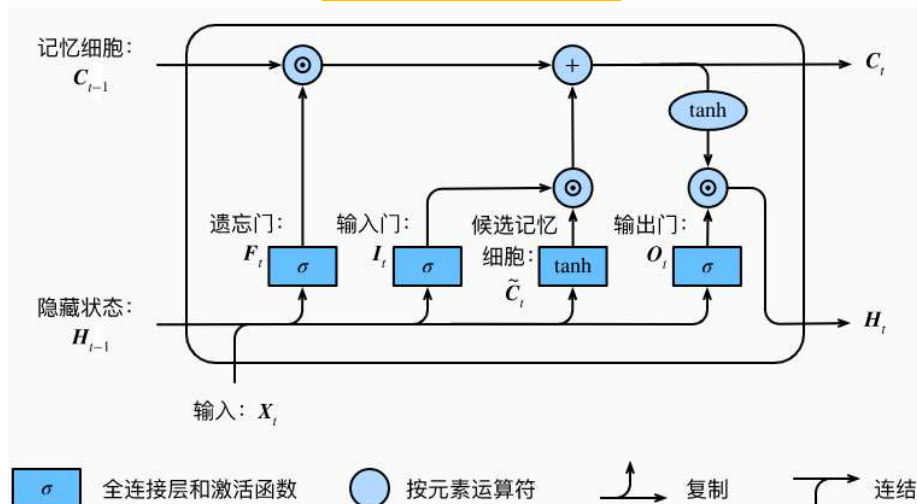
Sigmoid和tanh的导数都在 $[0,1]$ 范围内，随着时间序列的不断深入，小数的累乘就会导致梯度越来越小直到接近于0，这就是“梯度消失”现象。梯度消失就意味着消失那一层的参数再也不更新，那么那一层隐层就变成了单纯的映射层，毫无意义了。



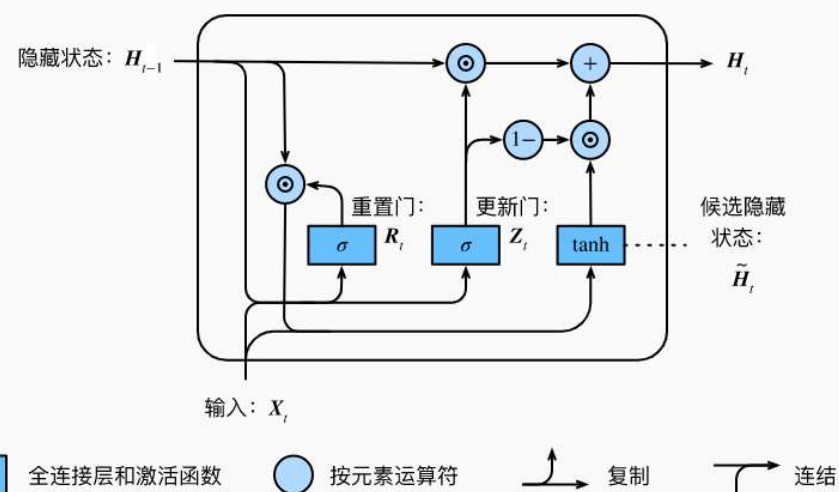
1 循环神经网络训练

- 如何解决？
 - 梯度爆炸：梯度削减
 - 梯度消失：
 1. 选取更好的激活函数
 2. 改变传播结构

LSTM



GRU



04

PART FOUR

小结和资源

Take Home Message and Resource



1

小结

- RNN的结构
- 语言模型
- Bi-RNN & Deep RNN
- RNN的训练
 - 随时间反向传播
 - 梯度爆炸
 - 梯度消失



1 小结

- Andrew Ng, Recurrent Neural Network

<https://www.bilibili.com/video/av88010803?p=1>

- 李宏毅，深度学习

<https://www.bilibili.com/video/BV1JE411g7XF>

http://speech.ee.ntu.edu.tw/~tlkagk/courses_ML20.html

